OCTOPUS: ONE BRAIN, MANY BRANCHES FOR LATENT-KV MULTI-AGENT REINFORCEMENT LEARNING

TECHNICAL REPORT

Tianming Sha
Stony Brook University
tianming.sha@stonybrook.edu

Junnan Li
University of Wisconsin–Madison
jli2786@wisc.edu

Siqi Zhao
University of Minnesota
zhao2052@umn.edu

ABSTRACT

Deep search and long-horizon reasoning naturally unfold as branching trajectories that pursue parallel sub-queries, abandon dead-end detours, and cross-check sibling findings. Yet modern serving engines, agent harnesses, and reinforcement learning (RL) trainers execute every episode as a flat, append-only text sequence: stretching a single context window leaves failed detours in working memory that dilute attention, in-place edits break prefix KV caching and recurrent states, and splicing caches across all layers overwrites hybrid recurrent working memory and triggers severe task hijacking.

We introduce **Octopus**, a co-designed serving runtime, GPU memory manager, and RL framework that executes episodes as a GPU-resident DAG over immutable radix cursors (89.6% GPU KV cache hit rate). Agents coordinate via five primitives: `fork` shares parent KV prefixes in $\mathcal{O}(1)$ with isolated 32K branch budgets, `rewind` restores hybrid recurrent checkpoints, `read_thread` mounts sibling KV pages in zero copy, and `wait` and `done` synchronize results. To prevent cross-branch interference in hybrid models, Octopus splices sibling KV pages exclusively into full-attention layers under monotonic position offsets while recurrent layers skip mounted spans, retaining 92.2% of full-text answer information and cutting off-task hijacking by $21\times$ (57.9% $\rightarrow$ 2.7%) with zero new KV pages. During RL, our prefix-tree packed 1D DAG kernel and **DAG-GRPO** objective deduplicate shared prefixes via prefix-deduplicating loss masks while keeping mounted KV edges differentiable in the forward–backward graph, cutting rollout–training KL divergence $32\times$ and routing reader gradients directly into producer KV projections. Trained on Qwen3.5-4B with binary outcome rewards, Octopus self-organizes a commander–worker hierarchy and peer-to-peer latent KV mesh from scratch, lifting BrowseComp-Plus accuracy under sparse BM25 from 26.0% (32K ReAct) and 32.1% (143K Single-Agent GRPO) to 49.6% (+17.5 points; 51.2% on OpenResearcher). Despite its 4B footprint, Octopus surpasses 9B context-editing RL (42.5%), matches OpenAI o3 under BM25 (49.3%), and outperforms dense-retrieval browser agents up to $250\times$ larger, including 30B Tongyi-DeepResearch (44.5%), 671B DeepSeek-R1 (16.4%), 1T Kimi-K2 (35.4%), GPT-4.1 (36.4%), and Claude 4 Opus (36.8%).

Code & Runtime: https://github.com/shatianming5/octopus

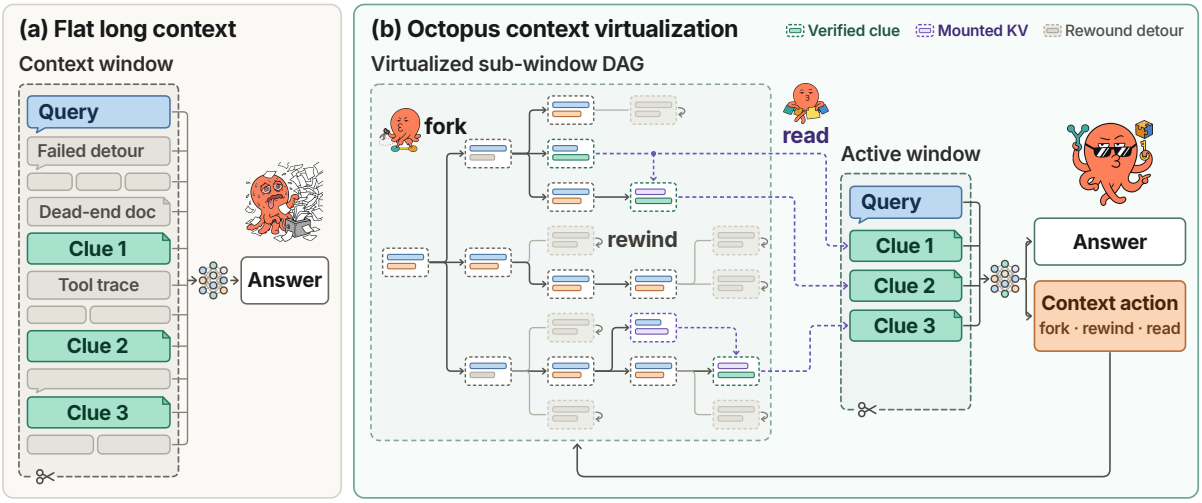


Figure 1: Flat append-only context vs. Octopus GPU-resident execution DAG. (a) Flat windows retain failed detours that dilute attention, while all-layer KV splicing corrupts hybrid recurrent states. **(b)** Octopus coordinates isolated 32K branches over shared GPU radix cursors via $\mathcal{O}(1)$ `fork`, `rewind`, and attention-only `read_thread` KV mounting.

1 Introduction

Deep search and multi-hop reasoning are inherently branching processes: an agent decomposes a question into parallel sub-queries, abandons dead-end detours, cross-checks sibling findings, and synthesizes verified evidence [1–3]. Yet modern serving engines, agent harnesses, and reinforcement learning (RL) trainers execute every episode as a flat, append-only text sequence [4–8]. Forcing tree-structured exploration onto a single linear context creates an immediate trilemma. A standard 32K window saturates after opening just a few 8K-token web pages, capping a 4B agent at 26.0% accuracy on BrowseComp-Plus (Table 3). Stretching that flat window to 143K under RL lifts accuracy to only 32.1% and stalls at 28.5% capacity (40.8K tokens), because failed detours remain in working memory and dilute attention over true evidence [9, 10]. Meanwhile, compacting or editing prior turns in place [11–15] discards exact identifiers, breaks append-only prefix KV caching, and invalidates sequential recurrent states in modern hybrid architectures such as Gated DeltaNet [16] and Mamba-2 [17].

Breaking out of a single flat window requires native branching execution, yet existing multi-agent, KV-sharing, and RL systems hit three structural walls. First, client-side multi-agent harnesses coordinate workers over stateless chat APIs, re-transmitting text histories and duplicating shared prefixes across sessions [5, 6, 18, 19]. Second, inference-time KV-sharing methods splice or re-rotate caches across all layers indiscriminately [20–23]; in hybrid recurrent-attention models, stepping recurrent states through a sibling's text or full-layer cache overwrites the reader's sequential working memory and hijacks over half of subsequent turns (55.2%–57.9%; Table 5). Third, existing branching RL frameworks exchange findings only as re-encoded text summaries [24–26], while naive sequence-level tree training either overcounts shared prefixes or severs cross-branch KV dependencies.

We introduce **Octopus** (Figure 1), a co-designed serving runtime, GPU memory manager, and RL framework that replaces the flat text tape with a GPU-resident execution DAG. Agents coordinate directly over immutable server-side radix cursors through five primitives: `fork` shares the parent's KV prefix in $\mathcal{O}(1)$ while granting each child a clean 32K growth budget; `rewind` rolls back dead-end detours and restores hybrid recurrent checkpoints $\mathbf{h}_k$; `read_thread` mounts sibling KV pages in zero copy; and `wait` and `done` synchronize results. Two co-designed mechanisms make latent cross-branch execution both hijack-free and end-to-end trainable. First, Octopus decouples sequential working memory from associative recall in hybrid models [27]: `read_thread` splices sibling KV pages exclusively into full-attention layers under a monotonic position offset, while recurrent layers skip mounted spans. This zero-copy mount allocates zero new KV pages, retains 92.2% of full-text answer information, and cuts off-task hijacking by 21× (57.9% → 2.7%). Second, our prefix-tree packed 1D DAG kernel and **DAG-GRPO** objective deduplicate shared prefixes by masking each tree segment to a single owner while keeping mounted KV edges differentiable in the forward–backward graph. Retaining live mount edges cuts rollout–training divergence by 32× ($\mathrm{KL} = 0.0142 \rightarrow 4.45 \times 10^{-4}$) and routes reader gradients directly into producer KV projections, keeping cross-branch latent communication strictly on-policy and end-to-end differentiable.

Trained on Qwen3.5-4B using pure binary outcome rewards with zero process shaping [28], Octopus discovers a self-organized division of labor from scratch (Figure 10). Without RL, exposing branching primitives yields no gain over 32K single-agent ReAct (26.3% vs. 26.0%). Under DAG-GRPO, the policy undergoes a sharp phase transition at Step 15: the root thread evolves into a dedicated commander that forks on Turn 1 in 98.0% of episodes (up 38×), offloads 92.8% of 8K-page reads and 91.5% of rewinds to sub-agents, and coordinates a peer-to-peer latent KV mesh averaging 23.9 zero-copy mounts per episode. Disabling `fork` at test time on the exact same trained weights collapses paired BrowseComp-Plus accuracy by 21.7 points (50.8% → 29.1%), proving that gains come from live collective execution rather than weight memorization.

Isolating noisy document reads across clean 32K branches keeps per-turn active prompts at 26.4K tokens while scaling explored context to 329.2K (199.0K net surviving), compressing critical-path length by 1.94×, more than doubling gold-document opening recall (21.1% → 43.3%), and lifting accuracy on the hardest multi-hop queries by 10× (3.4% → 34.1%). Overall, Octopus-4B under sparse BM25 reaches 51.2% on OpenResearcher and 49.6% on BrowseComp-Plus (+17.5 points over 143K Single-Agent GRPO; Figure 8, Table 3). Despite its 4B footprint, Octopus surpasses 9B context-editing RL (42.5%) [15], matches snippet-only OpenAI o3 under BM25 (49.3%), and outperforms dense-retrieval browser agents up to 250× larger—including 30B Tongyi-DeepResearch (44.5%), 671B DeepSeek-R1 (16.4%), 1T Kimi-K2 (35.4%), GPT-4.1 (36.4%), and Claude 4 Opus (36.8%). Our contributions are:

1. **GPU-Native Branching Execution Primitives (Sections 3.1 and 4.3):** We formulate long-horizon exploration as a GPU-resident execution DAG over five primitives (`fork`, `rewind`, `read_thread`, `wait`, and `done`) with isolated 32K branch budgets and bounded runtime admission.
2. **Hijack-Free Zero-Copy KV Mounting (Sections 3.2 and 4.2):** We splice sibling KV pages exclusively into full-attention layers under monotonic position offsets while recurrent layers skip mounted spans, retaining 92.2% answer information and cutting hijacking 21× (57.9% → 2.7%).

3. **Server-Resident Radix Cursors and Prefix-Tree Packed DAG Kernels (Sections 3.3 and 4.2):** We bind turns to immutable GPU radix cursors (89.6% KV cache hit rate) and compile prefix trees, recurrent checkpoints, and KV mounts into one 1D forward–backward pass.
4. **Latent-Communication Tree RL and Emergent Collective Scaling (Sections 3.4, 4.1 and 4.3):** We introduce DAG-GRPO with prefix-deduplicating loss masks and differentiable mount edges (32× lower rollout–training KL), eliciting emergent multi-agent division of labor and lifting Qwen3.5-4B to 49.6% on BrowseComp-Plus and 51.2% on OpenResearcher.

2 Related Work

Active Context Management, Compaction, and Branch-and-Return Folding. To operate within finite context budgets, recent agent harnesses periodically summarize history [11, 12], consolidate memory every turn [13, 29], or expose explicit compression and offloading tools [14, 30–33]. Context Language Models (CLM) paired with Suffix Cache Reuse (SCR) [15] allow in-place context edits, but mid-context mutations break prefix caching, force suffix copy and RoPE re-rotation, and invalidate sequential recurrent states in hybrid models. Closer to our setting, Context-Folding [25] folds sub-trajectories sequentially at depth 1, while Adaptive Parallel Reasoning (APR) [24] spawns parallel child threads under RL. However, both return compressed text summaries upon joining a branch—discarding exact identifiers and preventing lateral peer reads—and neither snapshots or rolls back hybrid recurrent states (Table 1). Octopus instead executes concurrent, unlimited-depth trees over shared GPU radix prefixes, supports exact recurrent-state backtracking, and shares sibling findings directly via zero-copy attention KV mounting.

Multi-Agent Harnesses and Cross-Agent KV Sharing. Existing multi-agent harnesses [5, 6, 18, 19] coordinate workers above stateless chat APIs, incurring repeated prompt re-transmission, re-tokenization, and quadratic KV duplication across sessions. To bypass text serialization, inference-time KV-sharing methods select attention layers in pure Transformers [34] or splice and RoPE-rerotate KV blocks across contexts [20–23, 35–37], while Cache-to-Cache [38] and DiffMAS [39] train supervised KV fusion modules on fixed graphs. Yet none of these methods support dynamic branching trees under on-policy RL, prevent recurrent-state corruption in hybrid architectures [16, 17, 27], or avoid RoPE phase inversion when a sibling branch runs longer than the reader. Octopus resolves all three limitations by splicing sibling KV pages exclusively into full-attention layers under rotation-free monotonic position offsets while recurrent layers skip mounted spans.

Reinforcement Learning for Long-Horizon and Branching Agents. Multi-turn agentic RL [1, 7, 8, 40–42] and stepwise context-editing RL [12, 15, 31] optimize flat trajectories segmented at turn or edit boundaries. Existing branching and hierarchical RL methods [24–26] exchange findings across branches only as re-encoded text summaries, whereas flattening trees into disconnected sequences either overcounts shared prefixes or drops cross-branch KV mounts. Octopus instead compiles prefix trees, recurrent-state rewinds, and live KV mount edges into one on-policy training graph ($\mathrm{KL} = 4.45 \times 10^{-4}$ vs. 0.0142 without mounts; Section 4.3), routing reader gradients directly into producer KV projections. To our knowledge, Octopus is the first on-policy RL system for agents that communicate through shared latent KV states, and the first on hybrid recurrent-attention models.

3 Method

Instead of re-sending growing chat transcripts on every turn, Octopus models a long-horizon episode as a cursor-addressed execution DAG $\mathcal{G} = (\mathcal{V}, \mathcal{E}_{\text{tree}}, \mathcal{E}_{\text{mount}})$. Analogous to an OS copy-on-write handle, each vertex $c \in \mathcal{V}$ is a lightweight cursor ID referencing an immutable snapshot of physical KV pages and recurrent state resident in GPU memory. Directed tree edges $(c \to c') \in \mathcal{E}_{\text{tree}}$ capture vertical state transitions (turn continuation, parallel `fork`, and `rewind`), whereas directed mount edges $(c_A \rightsquigarrow c_B) \in \mathcal{E}_{\text{mount}}$ capture horizontal coordination (`read_thread`), allowing branch B to attend directly to sibling A's physical KV pages. Octopus co-designs four layers around $\mathcal{G}$ (Figure 1): branching execution primitives (Section 3.1), layer-selective monotonic KV mounting (Section 3.2), server-resident radix cursors and prefix-tree packed DAG kernels (Section 3.3), and on-policy DAG-GRPO training (Section 3.4).

3.1 Branching Execution Primitives and Per-Branch Isolation

Existing single- and multi-agent harnesses sit above the serving engine, either enforcing a flat append-only loop or stitching stateless chat sessions at $\mathcal{O}(L)$ wire cost and $\mathcal{O}(N^2)$ KV duplication. Octopus instead co-designs the multi-agent harness with the GPU memory manager, running branches as lightweight GPU-resident threads across five primitives with isolated budgets and explicit termination guards (Figure 2, Table 2).

Table 1: Six-dimension architectural comparison between **Octopus** and five existing long-horizon agent paradigms (✓ supported, △ partial / degraded, ✗ unsupported / vulnerable).

System, Memory & RL Dimension	1. Long Window (Qwen-1M / Single-Agent RL)	2. Edit / Fold (CLM+SCR / Context-Folding)	3. Multi-Agent (AutoGen / CAMEL / APR)	4. Prior KV Sharing (Blend / Hogwild! / KVComm)	5. Sequence Agent RL (Search-R1 / RAGEN + Leaf-Split Trees)	Octopus (Ours) (Cursor + Attn Mount + DAG-GRPO)
1. Execution Topology & Isolation	✗ **Flat Linear** Detour pollution dilutes attention	△ **Sequential Fold** Depth-1 serial; no path speedup	✓ **Parallel Chats** Isolated sessions, no shared KV	✓ **Parallel Tree** Inference-only KV block sharing	✗ **Flat Sequences** Single-chain rollout or leaf-split tree	✓ **Parallel CoW Tree + DAG** Unlimited depth (32K/branch) + lateral peer `read`
2. Per-Turn Wire & Token Protocol	✗ $\mathcal{O}(L)$ **Resend** Full history resend & template drift	✗ **Re-Tokenize** Re-renders prompt after mid-edits	✗ $\mathcal{O}(NL)$ **Resend** Repeated text serialization	△ **Block Sync** External position & block tracker	✗ $\mathcal{O}(L)$ **Stitch** Re-tokenizes flat turns every step	✓ $\mathcal{O}(\Delta)$ **Server Cursor** Immutable radix cursor + startup template probe
3. Prefix KV & Hybrid State Reuse	△ **Append-Only** Invalidated upon any backtrack	✗ **Breaks Recurrence** Mid-edit / fold invalidates state	✗ $\mathcal{O}(N^2)$ **Isolated** 0% cross-agent KV cache reuse	✗ **Attn-Only / All-Layer** Corrupts hybrid recurrent state	✗ **Recomputes** $K\times$ Repeats shared trunk per leaf	✓ **Shared GPU Radix Tree** $\mathcal{O}(1)$ ref-counted KV prefix + cloned recurrent state
4. Cross-Branch KV Sharing & Hijack Safety	— Single-thread execution only	✗ **Lossy Text Return** Drops IDs; text pollutes memory	✗ $+L$ **Text Copy** Raw transcript hijacks reader	✗ $+L$ **Copy / Unshifted** RoPE copy pages or longer-sibling flip	— No cross-branch state reading	✓ **0-Copy Attn KV Mount** Recurrent layers skip mount + monotonic pos. offset
5. Backtrack Recovery & Tree Closure	✗ **No Backtrack** Dead-end docs remain in window	△ **KV Truncate Only** No recurrent snap or lateral `read`	✗ **No State Rewind** Cannot roll back branch KV or state	✗ **No State Snap** Cannot roll back recurrent state	✗ **No Branch Rewind** Append-only turn rollout buffer	✓ **Exact State Checkpoint** Reclaims branch budget + termination prompt guard
6. RL Kernel & Rollout Alignment	✗ **Detour Noise** Failed detours dilute credit	△ **Cut at Segments** No cross-branch KV mount edges	✗ **No Shared-KV RL** Independent chats or text-summary joins	✗ **Inference-Only** No packed tree training kernel	✗ **Drops KV Mounts** Sequence pass breaks rollout alignment	✓ **Prefix-Tree Packed Kernel** Prefix-deduplicating mask + differentiable KV mounts

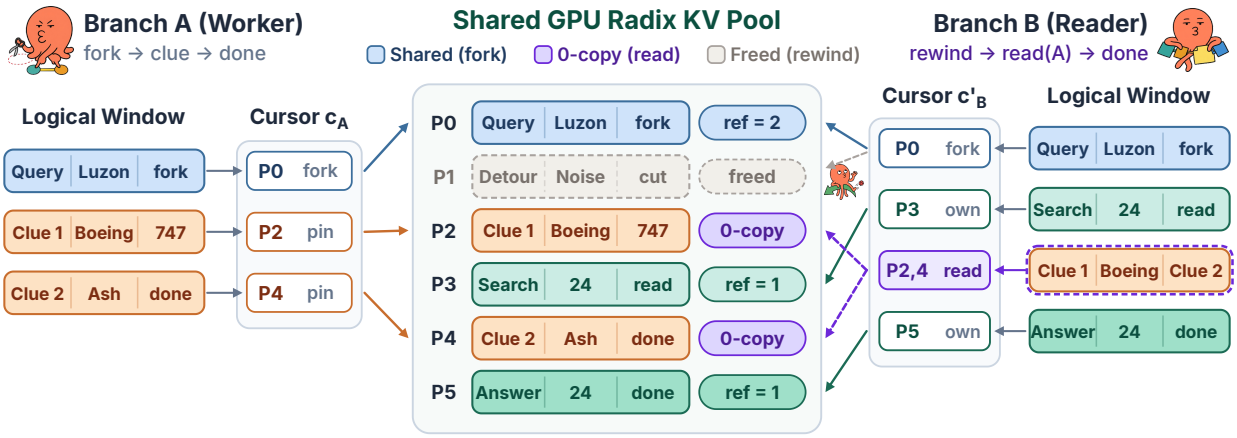


Figure 2: Cursor-addressed page translation and zero-copy KV page lending in Octopus (Sections 3.1 and 3.3). Worker and reader branches maintain isolated logical windows backed by a shared GPU radix KV pool: `fork` shares the parent prefix in $\mathcal{O}(1)$ (`ref = 2`), `rewind` reclaims dead-end pages, and `read` mounts pinned sibling KV pages in zero copy without allocating new physical blocks.

Orchestrator–Worker Coordination. An episode begins at a root orchestrator thread and expands recursively into concurrent worker branches. Calling `fork` at cursor c_{fork} spawns K parallel children in $\mathcal{O}(1)$ zero copy: all children share the parent's GPU KV prefix pages via reference counting ($+K$) while each receives a cloned recurrent state $\mathbf{h}_{\text{end}}(c_{\text{fork}})$. When a search query or 8K-token page open hits a dead end, `rewind` rolls the cursor back to step k, restores the exact recurrent checkpoint $\mathbf{h}_k$, reclaims steps $k+1 \dots t$ from the branch budget, and appends a short diagnostic `note`. For horizontal sharing, `read_thread` (abbreviated `read` in Table 2) mounts a completed step range $[a, b]$ from a sibling or cousin branch directly into the caller's full-attention layers without copying text or advancing recurrent states. Parents synchronize via `wait`, and every thread closes via `done`.

Per-Branch Budgets and Runtime Guardrails. Each thread operates under an independent growth budget $B_{\text{thread}} = 32{,}768$ tokens (with $R_{\text{ans}} = 2{,}048$ tokens reserved for final synthesis): only newly appended or generated tokens consume B_{thread}, whereas inherited parent prefixes and mounted sibling KV pages are shared in zero copy. To preserve prefix KV caching and prevent server saturation or unclosed rollouts, Octopus appends the `read_thread` tool schema only after a peer step completes, gates new episode admission when GPU KV pool usage exceeds $\tau_{\text{KV}} = 0.75$, and issues a single-turn loss-masked termination prompt ($m_{\ell,t} = 0$) if a thread emits plain text instead of `done` (Appendix A).

Table 2: Summary of the five execution primitives in Octopus (Section 3.1), their server-side cursor and KV-cache semantics, and their control-flow and budget semantics.

Primitive	Signature	Cursor & KV-Cache Semantics	Control-Flow & Budget Semantics
`fork`	`fork(branches={id: prompt})`	Clones parent cursor node in $\mathcal{O}(1)$; shares radix KV prefix; clones recurrent state $\mathbf{h}_{\text{end}}(c_{\text{fork}})$.	Spawns parallel child branches, each granted a fresh $B_{\text{thread}} = 32\text{K}$ token growth budget.
`read`	`read(branch, steps=[a, b])`	Zero-copy lends branch A's KV pages to attention layers $\mathcal{L}_{\text{mount}}$ w/ monotonic pos. (Equation (2)); recurrent layers skip span.	Injects a compact metadata stub (~ 15 tokens) into reader B's prompt without copying A's text.
`wait`	`wait(branches=[...])`	Suspends caller cursor until target child branches reach terminal `COMPLETED` state.	Returns brief branch status summaries and step manifests available for `read`.
`rewind`	`rewind(step=k, note)`	Restores cursor c_k and exact recurrent state snapshot $\mathbf{h}_k$ at step k; appends diagnostic `note`.	Reclaims dead-end token budget $(k+1 \dots t)$ while retaining the pruned branch in the training DAG.
`done`	`done(answer, evidence)`	Marks branch or root cursor `COMPLETED` and pins its KV pages in GPU radix pool for parent/sibling `read`.	Enforced by a 2K answer reserve and loss-masked termination prompt; returns verified answer and evidence.

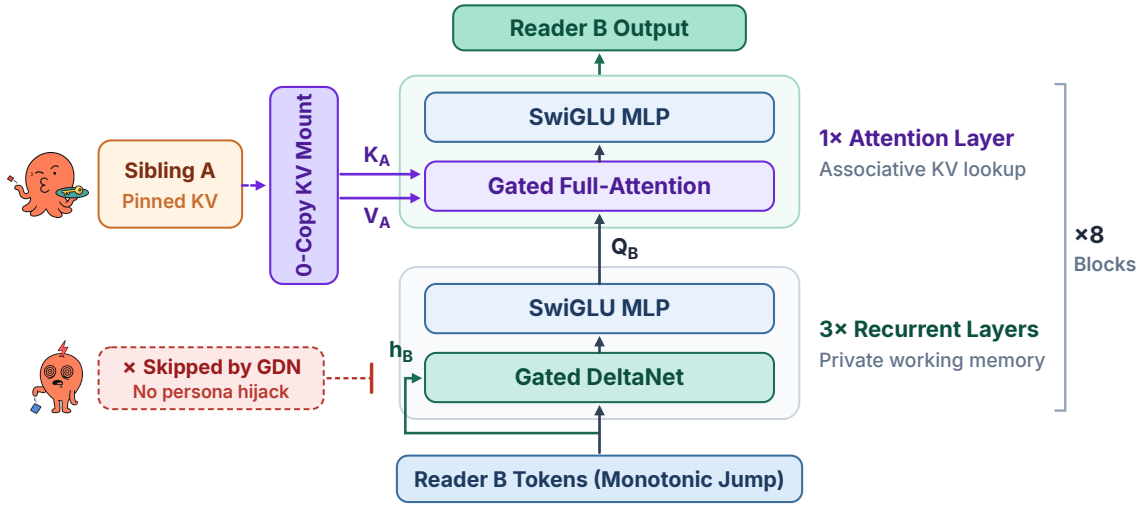


Figure 3: Layer-selective cross-branch KV mounting in hybrid recurrent-attention blocks (Section 3.2). Pinned sibling KV pages (K_A, V_A) are spliced in zero copy exclusively into full-attention layers for associative lookup, while recurrent Gated DeltaNet layers advance only the reader's private working memory $\mathbf{h}_B$ over post-mount tokens to prevent off-task hijacking.

3.2 Layer-Selective Monotonic KV Mounting

Sharing sibling findings via raw text injection or all-layer KV splicing causes reader branch B to abandon its own goal and pursue sibling A's task in over half of interactions (Section 4.2). We resolve this conflict by exploiting how recurrent and full-attention layers physically store history (Figure 3).

Decoupling Recurrent Working Memory from Associative Attention Lookup. Modern long-context backbones interleave linear recurrent layers (e.g., Gated DeltaNet or Mamba-2) with sparse full-attention layers [16, 17]. Crucially, a recurrent layer compresses history into a fixed-size rolling state $\mathbf{h}_B$ that is mutated by every incoming token, whereas a full-attention layer stores KV pages passively and reads them only when queried [27] (in pure-attention models, a sparse subset of middle retrieval layers plays an analogous role; Appendix C). Feeding sibling A's trajectory through recurrent layers directly overwrites $\mathbf{h}_B$—replacing B's active persona and goal with A's—whereas mounting A's KV pages into target full-attention layers $\mathcal{L}_{\text{mount}} \subseteq \mathcal{L}_{\text{attn}}$ exposes A's evidence as a passive lookup table. When reader B (at cursor c_B) mounts span $[s_A, e_A)$ from c_A, Octopus splices $c_A[s_A : e_A)$ exclusively into $\mathcal{L}_{\text{mount}}$ and skips $[s_A, e_A)$ in

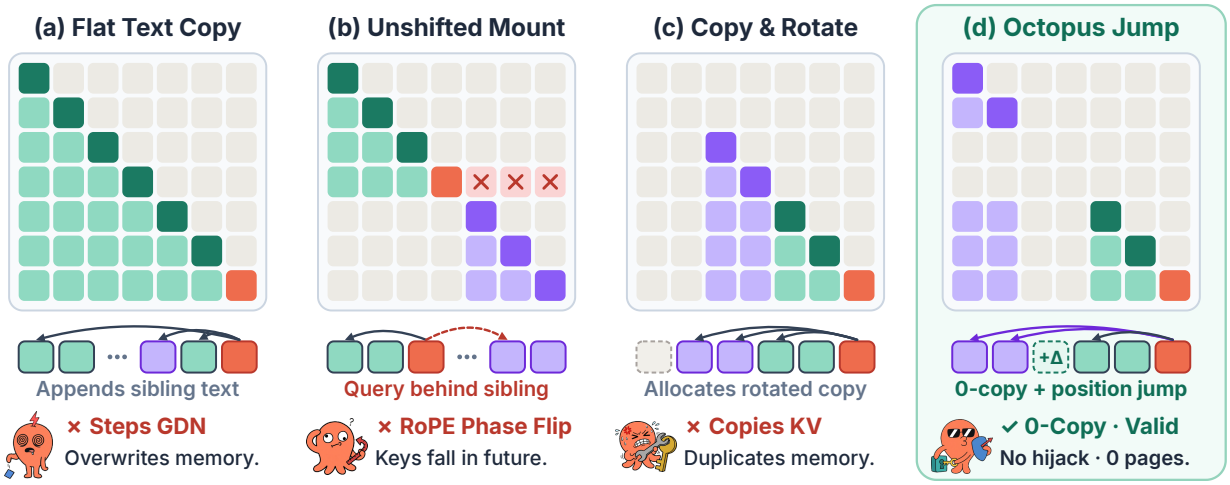


Figure 4: Comparison of cross-branch context sharing regimes (Sections 3.2 and 3.3): (a) flat text copying steps recurrent layers and causes task hijacking; (b) unshifted KV mounting inverts RoPE relative phases whenever sibling keys exceed the reader's position; (c) copying and re-rotating KV pages duplicates GPU memory per reader; (d) Octopus retains native zero-copy KV pages and jumps post-mount query positions monotonically under causal gathering.

all recurrent layers, advancing $\mathbf{h}_B$ solely over B's post-mount tokens Δ_B from $\mathbf{h}_{\text{end}}(c_B)$. Retrieved attention features on Δ_B still inform deeper recurrent layers ($l > l_{\text{attn}}$) via residuals without A's raw tokens overwriting $\mathbf{h}_B$.

Rotation-Free Monotonic Position Continuation. Splicing foreign KV pages into full-attention layers creates relative-position conflicts under 1D/M-RoPE (Figure 4), where attention scores depend on query–key phase differences:

$$\langle \mathcal{R}_q \mathbf{q},\, \mathcal{R}_k \mathbf{k} \rangle = \mathrm{Re}\left(\sum_d q_d k_d^* \, e^{i(q-k)\theta_d} \right). \tag{1}$$

Let $p_B = \mathrm{pos}_{\text{next}}(c_B)$ denote B's next unshifted position. Whenever sibling A has progressed further in token index than B ($e_A > p_B$), leaving B's queries at $q_B = p_B$ yields negative relative distances $q_B - k_A < 0$ for mounted keys $k_A \in (p_B, e_A)$, placing A's keys in B's causal future with inverted RoPE phases. Conversely, copying and RoPE-rerotating A's keys into B's local frame [21, 22, 34] duplicates GPU memory per reader. Octopus instead keeps mounted KV pages at their native positions and jumps B's post-mount tokens Δ_B monotonically past the highest mounted index (a monotonic position-continuation rule denoted **Rule K2**):

$$s(c'_B) = \max\Big(\mathrm{pos}_{\text{next}}(c_B),\ \max_{m \in \mathrm{Mounts}(c'_B)} \big(\mathrm{pos}_{\text{end}}(m) + 1 \big) \Big). \tag{2}$$

For example, if B is at position 2,000 and mounts A's tokens 1,500–4,000, B's next token jumps to 4,001: every visible key now lies strictly in B's past ($q_B - k \geq 1$) with zero KV copying, while position-free recurrent and convolution layers are unaffected. Although mounted indices 1,500–1,999 overlap with B's pre-mount tokens counted from the fork point, content-based attention keys disambiguate colliding positions without re-rotation (Section 4.2).

3.3 Server-Resident Radix Cursors and Prefix-Tree Packed DAG Kernels

Inference Engine: Server-Resident Radix Cursors. To eliminate $\mathcal{O}(L)$ prompt re-transmission across turns and forks, the serving engine stores each completed turn as an immutable cursor c backed by a GPU radix prefix tree of reference-counted KV pages. Any continuation, `fork` child, or `rewind` recovery passes cursor ID c and incremental messages Δm, extending the sequence in $\mathcal{O}(|\Delta m|)$ time and wire bytes:

$$\mathbf{x}(c') = \underbrace{\big(\mathbf{x}_{\text{prompt}}(c) \,\oplus\, \mathbf{x}_{\text{out}}^{\setminus\text{stop}}(c) \,\oplus\, \tau_{\text{term}}(c) \big)}_{\text{Immutable Server-Resident Cursor } \mathbf{x}(c)} \oplus \mathrm{Render}_{\Delta}\big(\mathcal{T},\ \Delta m,\ \mathrm{ctx}(c) \big), \tag{3}$$

where $\mathbf{x}_{\text{out}}^{\setminus\text{stop}}(c)$ strips trailing stop tokens and $\tau_{\text{term}}(c)$ terminates the assistant turn. Because modern reasoning chat templates $\mathcal{T}$ are often non-append-only—for instance, stripping earlier `<think>` blocks on new user turns—a one-time (~ 0.1 s) startup probe extracts an incremental template slicer Render_{Δ} and encodes runtime notes as tool responses, achieving bit-exact tokenization across seven model families (Appendix A). Mounted KV spans are pinned via per-page reference counts and transparently re-prefilled from cursor metadata if evicted.

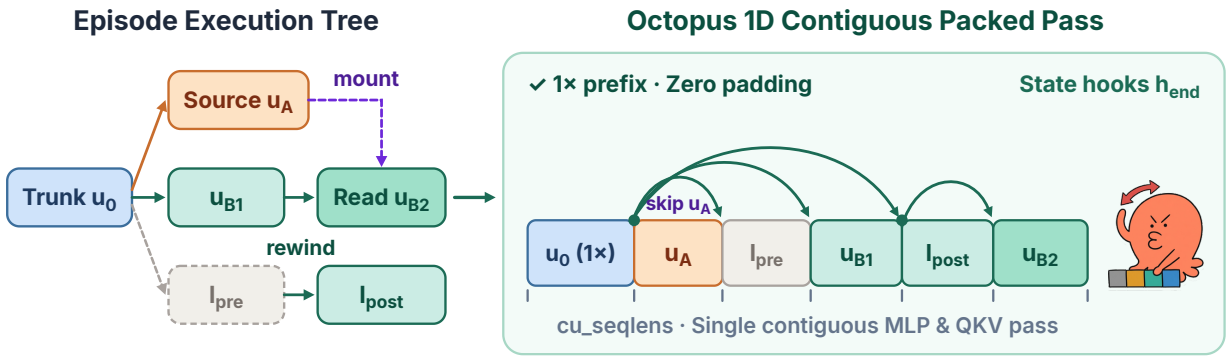


Figure 5: Tree-to-1D contiguous sequence packing (Section 3.3). Octopus deduplicates each tree segment once into an unpadded 1D strip and passes recurrent states $\mathbf{h}_{\text{end}}(p(u))$ across segment boundaries while skipping mounted sibling spans.

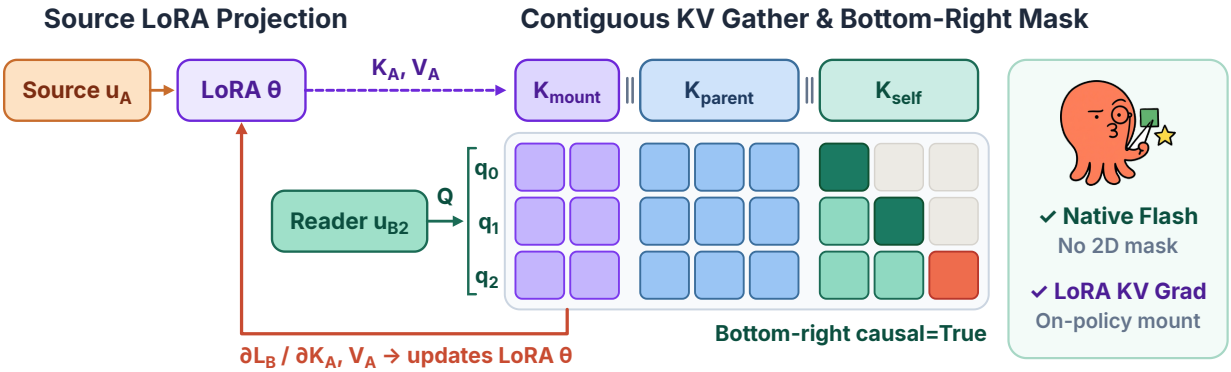


Figure 6: Contiguous KV gathering and cross-branch LoRA backpropagation (Sections 3.3 and 3.4). Concatenating mounted sibling keys, ancestor prefixes, and active reader tokens places the causal diagonal at the bottom-right corner for native FlashAttention (`causal=True`), while routing $\partial \mathcal{L}_{u_B} / \partial \mathbf{K}_{u_A}, \mathbf{V}_{u_A}$ directly into source branch A's LoRA adapter θ.

Training Engine: Tree-to-1D Contiguous Packed DAG Kernels. Splitting a branched episode into separate leaf sequences during RL either recomputes shared trunks K times or drops mounted KV spans. Octopus instead compiles each episode DAG into one prefix-tree packed forward–backward pass (Figure 5):

1. **Node Splitting and Prefix Deduplication:** At every `fork`, `rewind`, or `read_thread` boundary, Octopus partitions the episode into maximal contiguous token nodes $u = \big(p(u),\ \mathcal{M}(u),\ s(u)\big)$, recording tree parent $p(u)$, active mounted segments $\mathcal{M}(u)$ (including transitive mounts), and monotonic start position $s(u)$ (Equation (2)). Each unique node u is evaluated once regardless of how many branches or rewinds descend from it.
2. **1D Contiguous MLP and Projection Pass:** Deduplicated nodes $\{u_1, \ldots, u_M\}$ are concatenated topologically into one unpadded 1D tensor (up to 262K tokens) and run through all MLPs and QKV projections in one matrix pass.
3. **Dual-Mechanism Graph Routing:** Across the packed 1D strip, recurrent and attention layers route activations along $\mathcal{G}$ via two complementary hooks. On recurrent layers (Gated DeltaNet and Mamba-2), boundary hooks initialize each node u from its parent's terminal state $\mathbf{h}_{\text{end}}(p(u))$ (plus the last $w-1=3$ causal-convolution tokens), accumulate $\partial \mathcal{L} / \partial \mathbf{h}$ back to $p(u)$ in backward, and skip $\mathcal{M}(u)$ (Figure 5). On each mounted full-attention layer $l \in \mathcal{L}_{\text{mount}}$, node u gathers precomputed K/V tensors as (Figure 6):

$$\mathbf{K}_u^{(l)} = \Big[\bigoplus_{m \in \mathcal{M}(u)} \mathbf{K}_m^{(l)} \,\Big\|\, \mathbf{K}_{p(u)}^{(l)} \,\Big\|\, \mathbf{K}_{\text{self}(u)}^{(l)} \Big], \qquad \forall l \in \mathcal{L}_{\text{mount}}. \tag{4}$$

Placing $\text{self}(u)$ at the rightmost tail exploits rectangular FlashAttention (`causal=True`; Figure 4(d), Figure 6): whenever $|\mathbf{Q}_u| < |\mathbf{K}_u|$, FlashAttention aligns the causal triangle to the rightmost $|\mathbf{Q}_u|$ keys ($\text{self}(u)$) while leaving all preceding mounted and ancestor keys unmasked, avoiding custom 2D block-sparse masks and matching SGLang rollout log-probabilities within $|\Delta \log \pi_\theta| = 0.0038$.

3.4 DAG-GRPO Training with Prefix-Deduplicating Masks and Differentiable KV Mounts

Prefix-Deduplicating Loss Mask and RLOO Advantage. For each prompt, Octopus samples G branched episode trees $\{\tau_1, \ldots, \tau_G\}$ and scores each terminal outcome $y(\tau_i) \in [0, 1]$ with binary reward $R(\tau_i) = 2y(\tau_i) - 1$. Following REINFORCE Leave-One-Out (RLOO) [43], we compute a critic-free group advantage without standard-deviation normalization (Figure 7):

$$\hat{A}(\tau_i) = R(\tau_i) - \frac{1}{G-1} \sum_{j \neq i} R(\tau_j), \tag{5}$$

and broadcast $\hat{A}(\tau_i)$ across all threads of τ_i, including rewound leaves so the `rewind` decision itself receives outcome credit. Standard sequence-RL objectives sum token losses along root-to-leaf trajectories $\{\ell \in \tau_i\}$, which on a tree naively weights each shared trunk node u by its number of descendant leaves. To map trajectory-level RL directly onto our prefix-tree packed pass (Section 3.3), we assign each node u to the unique leaf $\ell = \text{owner}(u)$ that first generated it via a prefix-deduplicating ownership mask over action tokens $\mathcal{V}_{\text{gen}}(\tau_i)$:

$$m_{\ell,t} = \mathbb{I}\Big[\ell = \text{owner}(u_{\ell,t}) \ \wedge \ x_{\ell,t} \in \mathcal{V}_{\text{gen}}(\tau_i)\Big], \tag{6}$$

setting $m_{\ell,t} = 1$ once per generated token on the packed 1D strip and $m_{\ell,t} = 0$ on prompts, tool outputs, inherited or pre-rewind prefixes ($t \leq k$), and mounted sibling spans $\mathcal{M}(u)$. Summing with $m_{\ell,t}$ recovers the exact DAG score function:

$$\sum_{\ell \in \tau_i} \sum_{t=1}^{|\ell|} m_{\ell,t} \nabla_\theta \log \pi_\theta(x_{\ell,t} \mid \mathcal{G}_{\ell,<t}) = \nabla_\theta \log \mathbb{P}_\theta(\tau_i). \tag{7}$$

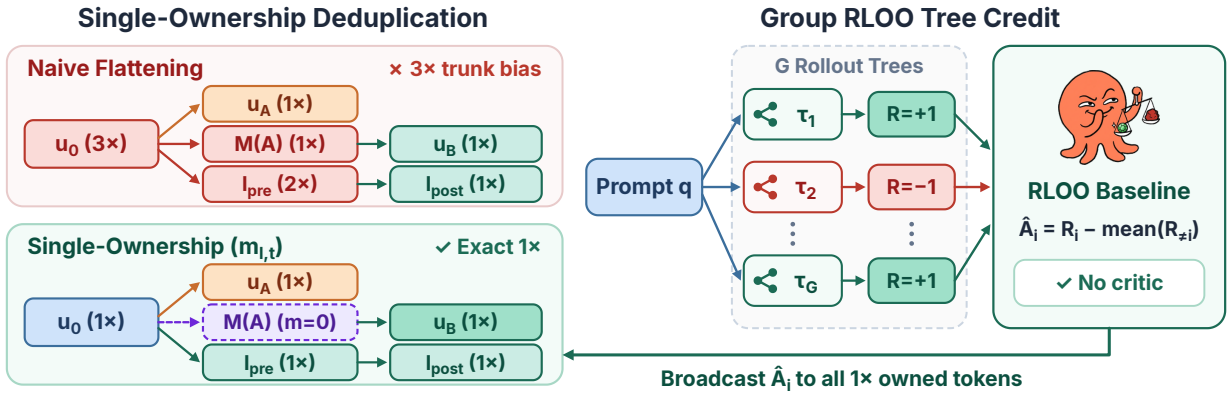


Figure 7: Prefix-deduplicating loss masking and Group RLOO tree credit assignment in DAG-GRPO (Section 3.4). Ownership masking ($m_{\ell,t} \in \{0, 1\}$) counts every generated tree segment once while masking mounted spans, and broadcasts critic-free Leave-One-Out group advantages $\hat{A}(\tau_i)$ across all owned tokens.

Differentiable On-Policy Mount Edges and Clipped Objective. During the prefix-tree packed pass, source node u_A's keys and values at each mounted layer $l \in \mathcal{L}_{\text{mount}}$ are projected through trainable LoRA weights $\theta = \{\mathbf{B}_\theta, \mathbf{A}_\theta\}$:

$$\mathbf{K}^{(l)}_{u_A}(\theta) = \big(\mathbf{W}^{(l)}_K + \mathbf{B}^{(l)}_{K,\theta}\mathbf{A}^{(l)}_{K,\theta}\big)\mathbf{H}^{(l)}_{u_A}(\theta), \qquad \mathbf{V}^{(l)}_{u_A}(\theta) = \big(\mathbf{W}^{(l)}_V + \mathbf{B}^{(l)}_{V,\theta}\mathbf{A}^{(l)}_{V,\theta}\big)\mathbf{H}^{(l)}_{u_A}(\theta), \tag{8}$$

and gathered directly by reader node u_B in Equation (4) without detaching the mount edge. Mounted spans carry loss mask $m_{\ell,t} = 0$ so A's tokens are never double-counted as B's actions, while attention backpropagation routes $\partial \mathcal{L}_{u_B}/\partial \mathbf{K}^{(l)}_{u_A}$ and $\partial \mathcal{L}_{u_B}/\partial \mathbf{V}^{(l)}_{u_A}$ directly into A's LoRA weights (Figure 6). Unlike text-summary branching—which trains θ only on re-encoded summary tokens—keeping live mounts in the packed graph preserves on-policy rollout alignment ($\text{KL} = 4.45 \times 10^{-4}$ vs. 0.0142 when dropped; Section 4.3) and lets reader B's loss directly shape how θ encodes producer A's KV state. We optimize the clipped DAG-GRPO objective:

$$\mathcal{J}_{\text{DAG-GRPO}}(\theta) = \mathbb{E}_{\{\tau_i\}_{i=1}^{G}}\left[\frac{1}{G}\sum_{i=1}^{G}\frac{1}{Z_i}\sum_{\ell \in \tau_i}\sum_{t=1}^{|\ell|} m_{\ell,t}\, \min\Big(r_{\ell,t}(\theta)\hat{A}(\tau_i),\ \text{clip}\big(r_{\ell,t}(\theta), 1-\epsilon, 1+\epsilon\big)\hat{A}(\tau_i)\Big)\right], \tag{9}$$

where $Z_i = \sum_{\ell \in \tau_i}\sum_{t=1}^{|\ell|} m_{\ell,t}$ counts owned generated tokens in τ_i and $r_{\ell,t}(\theta) = \pi_\theta(x_{\ell,t} \mid \mathcal{G}_{\ell,<t})/\pi_{\theta_{\text{old}}}(x_{\ell,t} \mid \mathcal{G}_{\ell,<t})$. Normalizing by $1/Z_i$ stabilizes gradient magnitudes across trees spanning 1K to 199K tokens, though analogous to Dr. GRPO [44] it introduces a tree-size length weighting that softens per-token penalties on failed large trees (Section 5).

4 Experiments

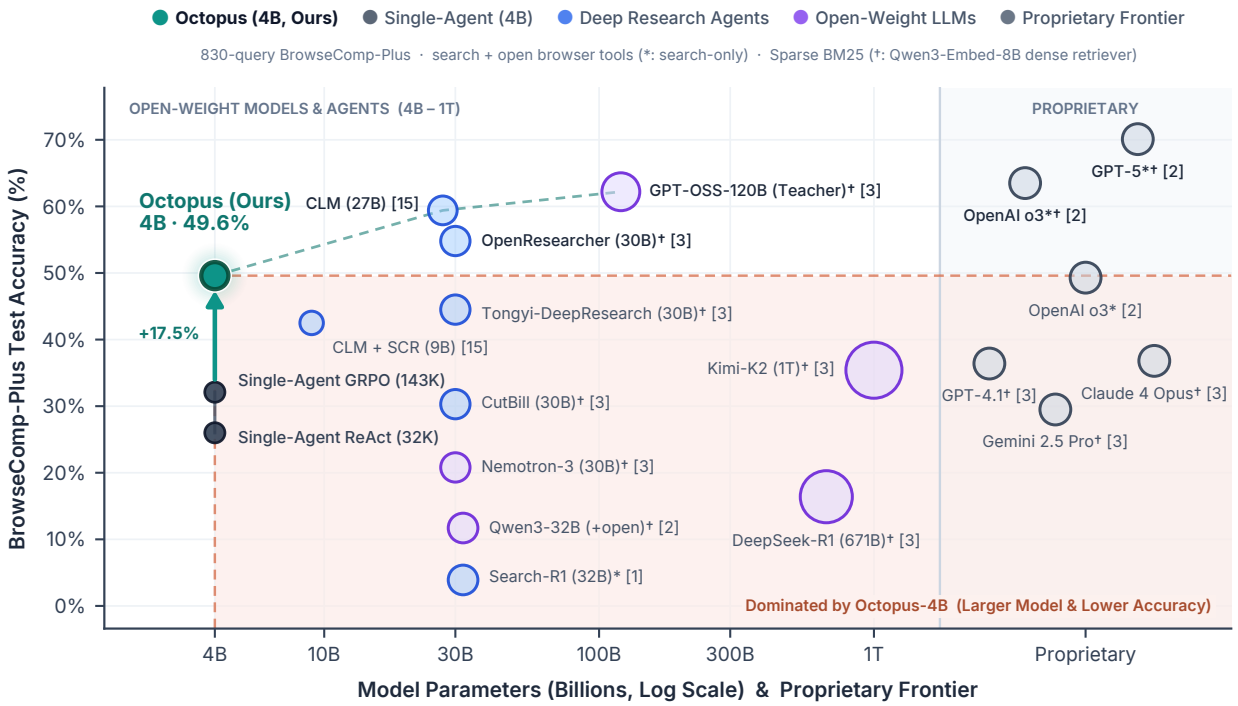


Figure 8: BrowseComp-Plus test accuracy (830 queries) vs. model parameter scale (4B–1T log scale and proprietary models) [2]. Systems use browser reading tools (`search+open[+find]`) over sparse BM25 unless marked * (`search`-only) or † (`Qwen3-Embed-8B` dense retrieval). **Octopus (4B)** reaches **49.6**% under BM25, dominating larger open-weight and proprietary systems across the shaded region.

We evaluate Octopus around three questions: *(i)* ***Deep Search Performance (Section 4.1)***: How does a 4B branching agent trained with Octopus compare against proprietary frontier models, 9B–1T open-weight systems, and controlled 4B single-agent baselines (Figure 8, Table 3)? *(ii)* ***Context and KV-Sharing Efficiency (Section 4.2)***: Does branch-isolated 32K exploration convert context into gold evidence and reuse GPU prefix caches better than flat 143K windows or in-place edits, and does attention-only KV mounting prevent off-task hijacking (Figure 9, Tables 4 and 5)? *(iii)* ***Multi-Agent Scalability (Section 4.3)***: How does accuracy scale with deployed agent count and unlimited tree depth ($D_{\max} = \infty$), and what collective behaviors emerge under pure outcome RL (Figure 10, Table 6)?

4.1 End-to-End Deep Search Performance

Experimental Setup, Corpus Isolation, and Compute Accounting. We train Qwen3.5-4B (rank-64 LoRA, $B_{\text{thread}} = 32{,}768$ with 2,048-token answer reserve; $G = 8$, batch size 16) on 3,040 OpenResearcher queries (500 held out for Validation, stratified by teacher pass rate from the 6,102-query MiroVerse pool [3]) over the 14.9M-document `OpenResearcher-Corpus`, and evaluate zero-shot on BrowseComp-Plus (830 queries) over its separate 100,195-document fixed corpus [2]. Cross-dataset auditing confirms 0/3,040 exact or 13-gram question overlap (max token Jaccard 0.265) and only 0.02% (2/8,306) training gold URL overlap with BrowseComp-Plus gold documents (two general Wikipedia pages with zero shared answers, leaving test accuracy unchanged at 49.6% when excluded). Both corpora use sparse BM25 (10×512-token `search` hits, vs. $12\times$ higher-recall dense† and snippet-only* baselines in Figure 8 and Table 3) and paginated 8K-token `open`, graded by Qwen3.5-27B at $T = 0$ [15]. Following Shao et al. [15], we report prefix-reuse inference PFLOPs per query (PF/Q; 89.59% GPU KV hit rate, 8-layer `read_thread`) and critical-path vs. total completion tokens (**Crit.** C **/ Total**).

Controlled Attribution and Parameter–Accuracy Frontier. On Qwen3.5-4B under sparse BM25 (Table 3), single-agent ReAct saturates its 32K window within five to six turns and plateaus at 24.6%–26.0%. Stretching one flat window to 143K under GRPO (**Single-Agent GRPO**) reaches only 32.1% (43.2% Validation), stalling after 40.8K tokens (28.5% utilization) as unpruned retrieval noise pollutes working memory. Exposing branching tools without RL (**Multi-Agent Fork, 0-Shot**) stays flat at 26.3% (0.41 branches/episode). Training depth-1 teams with DAG-GRPO lifts

Table 3: End-to-end deep search results on **BrowseComp-Plus** (830 test queries) [2] and **OpenResearcher Validation** (500 queries) [3]. Columns report prefix-reuse inference PFLOPs (**PF/Q**), accuracy under sparse BM25 (†dense retrieval, *`search`-only), evidence recall, and critical-path vs. total completion tokens (**Crit.** C **/ Total**). **Octopus (4B)** lifts BrowseComp-Plus accuracy by $+17.5$ points over 143K Single-Agent GRPO.

Method / Agent System	Backbone / Scale	Tools	PF/Q ↓	OR Val Val Acc. ↑	BrowseComp-Plus Official Test Benchmark (830 Queries) Test Acc. ↑ BM25 / Dense†	Search Rec. ↑	Open Rec. ↑	Gold Rec. ↑	Context Used / Cap	Crit. C / Total
Proprietary Frontier Models on BrowseComp-Plus [2, 3]										
Claude Sonnet 4 [2]	Proprietary	`s-only`*	—	—	14.3%* (36.8%*†)	21.3%*	—	—	≤ 200K	1.00×
Gemini 2.5 Pro [2, 3]	Proprietary	`s+o+f`	—	—	29.5%† (19.0%*)	22.8%*	—	—	≤ 1M	1.00×
GPT-4.1 [2, 3]◇	Proprietary	`s+o+f`	—	—	36.4%† (14.6%*)	16.4%*	—	—	≤ 128K	1.00×
Claude 4 Opus [2, 3]	Proprietary	`s+o+f`	—	—	36.8%† (15.5%*)	23.0%*	—	—	≤ 200K	1.00×
OpenAI o3 [2]	Proprietary	`s-only`*	—	—	49.3%* (63.5%*†)	56.6%*	—	—	≤ 200K	1.00×
GPT-5 [2]	Proprietary	`s-only`*	—	—	55.9%* (70.1%*†)	61.7%*	—	—	≤ 256K	1.00×
Open-Weight Flagship LLMs & Specialized Deep Research Agents (9B–1T Backbones)										
Search-R1-32B (RL) [1, 2]	32B Dense	`s-only`*	0.46	—	3.9%* (10.4%*†)	2.6%*	—	—	≤ 32.0K	1.00×
Qwen3-32B (+open) [2]	32B Dense	`s+o`	0.36*	—	11.7%† (3.5%*)	3.1%*	—	—	≤ 32.0K	1.00×
DeepSeek-R1 [3, 28]	671B-A37B	`s+o+f`	—	—	16.4%†	—	—	—	≤ 128K	1.00×
Nemotron-3-Nano [3]	30B-A3B	`s+o+f`	—	—	20.8%†	—	—	—	≤ 128K	1.00×
CutBill [3]	30B-A3B	`s+o+f`	—	—	30.3%†	—	—	—	≤ 128K	1.00×
Kimi-K2 [3]	1T-A32B	`s+o+f`	—	—	35.4%†	—	—	—	≤ 128K	1.00×
Context-Summary (0-Shot → RL) [15]	Qwen3.5-9B	`s+o+sum`	4.01→2.19	—	34.7→42.1%	—	—	—	≤ 32.0K	1.00×
CLM + SCR (0-Shot → RL) [15]	Qwen3.5-9B	`s+o+ret`	1.52→1.34	—	28.8→42.5%	—	—	—	≤ 32.0K	1.00×
Tongyi-DeepResearch [3]	30B-A3B	`s+o+f`	—	—	44.5%†	—	—	—	≤ 128K	1.00×
OpenResearcher (SFT) [3]	30B-A3B	`s+o+f`	—	—	54.8%†	—	—	—	≤ 128K	1.00×
CLM [15]‡	Qwen3.6-27B	`s+o+ret`	5.40	—	59.4%	—	—	—	≤ 32.0K	1.00×
GPT-OSS-120B (Teacher) [2, 3]	120B MoE	`s+o(+f)`	—	—	56.4 / 62.2%† (28.7%*)	35.5%*	—	—	≤ 128K	1.00×
Controlled Comparison Arms on Qwen3.5-4B (BM25 Index, Rank-64 LoRA, $B_{\text{thread}} = 32{,}768$ Per-Thread Cap)										
Single-Agent ReAct (100-Turn Cap)	Qwen3.5-4B	`s+o`	0.28	40.0%	24.6%	25.9%	6.4%	9.7%	28.6K /32K	1,097 / 1,097
Single-Agent ReAct (32K Window)	Qwen3.5-4B	`s+o`	0.33	40.0%	26.0%	28.6%	7.5%	12.2%	32.6K /32K	1,043 / 1,043
Multi-Agent Fork (0-Shot, 0.41 br)	Qwen3.5-4B	`s+o+tree`	0.38	41.6%	26.3% (24.8%)	30.5%	8.6%	12.5%	38.1K	1,590 / 1,811
Single-Agent GRPO (143K Window)	Qwen3.5-4B	`s+o`	0.43	43.2%	32.1%	30.5%	14.6%	21.1%	40.8K /143K	4,075 / 4,075
Octopus (Process-Gated RL, $D_{\max}$=1)	Qwen3.5-4B	`s+o+tree`	0.93	45.2%	39.5%	—	—	—	98.4K	4,877 / 9,120
Octopus (Pure Outcome, $D_{\max}$=1)	Qwen3.5-4B	`s+o+tree`	1.57 (0.70§)	**51.2%**§ **(+8.0)**	42.6% **(+10.5)**	49.5%	29.0%	37.7%	143.5K	9,918 / 18,418
Octopus (Ours, Recursive $D_{\max}$=∞) DAG-GRPO (6.41 **br**, 1-**Epoch**§)	**Qwen3.5-4B**	`s+o+tree`	**2.00**	**46.6%**§ **(Step 20)**	**49.6% (+17.5)**	**58.2%**	**36.0%**	**43.3% (+22.2)**	**199.0K**¶ **(329K expl.)**	**9,919 / 19,251**

Tools: `s=search`, `o=paginated open`, `f=find`, `sum/ret=summary/retention`, `tree=fork/rewind/read_thread`. **Rec.** columns report per-query recall over evidence (6.1/query) and gold (2.9/query) docs (58.9% open ≥ 1 gold doc vs. 20.7% 0-shot). †`Qwen3-Embed-8B` dense retrieval (14.5% vs. 1.2% BM25 Recall@5 [2]); GPT-OSS-120B: 56.4% (`s+o`) / 62.2%† (`s+o+f`) [3], 28.7%* / 42.9%*† [2]. ◇GPT-4.1: 43.6%† with full `get_document`. ‡On 150 dense queries, Context-Folding scores 62.0% vs. 44.6% ReAct+GRPO (Seed-OSS-36B [25]). §$D_{\max} = 1$: Step 35 scores 42.6% BCP (46.5% Val, 1.57 PF/Q), Step 55 scores 51.2% Val (0.70 PF/Q); $D_{\max} = \infty$ ($W_{\max} = 10$, Step 45 = 1 epoch): 49.6% BCP (95% CI [46.2%, 53.0%]; 49.3% across all 830 queries), 46.6% Val at Step 20. ¶Net post-`rewind` context (329.2K explored).

BrowseComp-Plus accuracy to 39.5% when gated by heuristic process penalties (**Process-Gated RL**) and 42.6% under pure binary outcome credit (+10.5 points; 51.2% on OpenResearcher Validation at 0.70 PF/Q). Removing the depth cap for a single epoch ($D_{\max} = \infty, W_{\max} = 10$) reaches 49.6% on BrowseComp-Plus (49.3% strict /830)—gaining +17.5 points over 143K Single-Agent GRPO and +23.3 paired points over the 0-shot base policy at 2.00 PF/Q.

Across model scales (Figure 8), our 4B BM25 agent outperforms 2.25× larger 9B context-editing RL agents on the identical BM25 harness by +7.1–7.5 points (42.1% Context-Summary+RL at 2.19 PF/Q; 42.5% CLM+SCR at 1.34 PF/Q) [15] and edges past OpenAI o3 under BM25 (49.3%*). Moreover, Octopus-4B under sparse BM25 surpasses much larger open-weight and proprietary agents equipped with 12× higher-recall dense retrieval (†), including 671B DeepSeek-R1 (16.4%†), 1M-window Gemini 2.5 Pro (29.5%†), 1T Kimi-K2 (35.4%†), Claude 4 Opus (36.8%†), and 30B Tongyi-DeepResearch (44.5%†). Only 7×–30× larger specialized open-weight systems (27B–120B, 54.8%–62.2%) and dense-retriever o3/GPT-5 (63.5%*†–70.1%*†) score higher, placing Octopus-4B atop the sub-27B frontier.

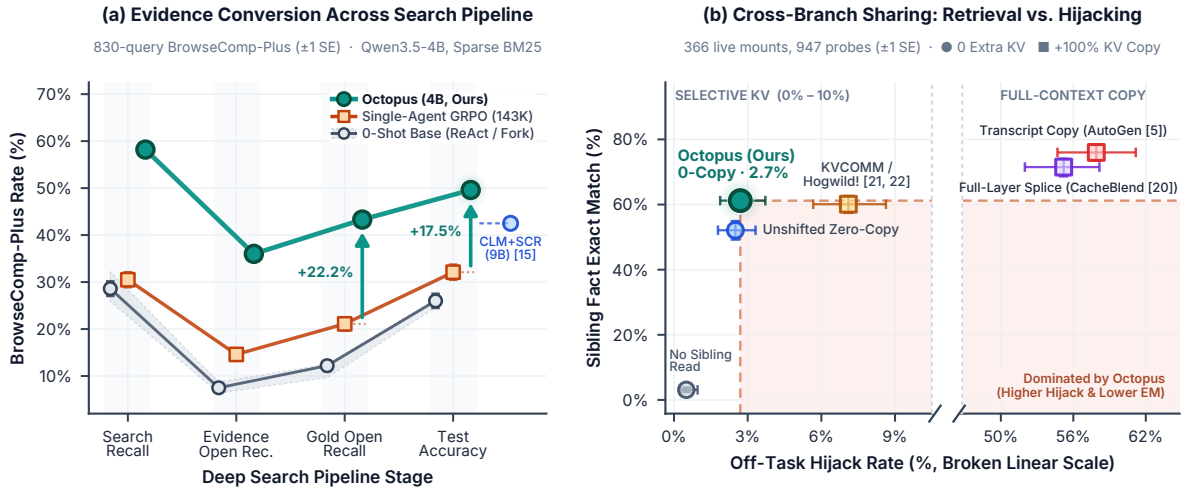


Figure 9: Context conversion efficiency and cross-branch KV sharing on Qwen3.5-4B (±1 SE). **(a)** Four-stage BrowseComp-Plus recall-to-accuracy pipeline (830 queries; per-query recall over 6.1 evidence / 2.9 gold docs, 58.9% opening $\geq$ 1 gold doc). **(b)** Sibling exact match vs. task hijacking across 366 live `read_thread` mounts (947 probes): **Octopus** retains 92.2% answer information at 2.7% hijacking with 0 extra KV pages.

4.2 Context Utilization and Compute Efficiency

Branch-Isolated Context Conversion, Evidence Grounding, and Radix Prefix Reuse. Figure 9(a) and Table 4 trace context conversion across the four BrowseComp-Plus stages (search, evidence-open, and gold-open recall, plus accuracy). Both 32K ReAct and 0-shot forking stall at 9.7%–12.5% gold-open recall (20.7% opening $\geq$ 1 gold doc). Stretching a flat window to 143K under GRPO fails to unlock deeper search: with 0% pruning of failed detours and noisy 8K pages, generated tokens bloat 3.9× (1,043 $\rightarrow$ 4,075) while context usage stalls at 40.8K/143K (28.5% capacity, 21.1% gold-open recall).

By contrast, Octopus isolates sub-queries into clean 32K branches and prunes 38.8% of post-initial steps via `rewind` (6.99 rewinds/episode), keeping active prompts at 26.4K tokens while exploring 329.2K (12.5× active; 541.3K on $\geq$ 250-call queries) and retaining 199.0K net tokens. Parallel execution compresses 19,251 generated tokens into a 9,919-token critical path (1.94×), lifting fractional gold-open recall from 21.1% to 43.3% (58.2% search and 36.0% evidence-open recall). Because BrowseComp-Plus annotates 2.89 gold documents per query (67.2% multi-gold) and episodes terminate once an opened gold page verifies the answer, Octopus opens at least one gold document on 58.9% of queries (81.9% conditional accuracy vs. 3.5% otherwise). Among correct predictions, 97.1% open at least one gold document and 2.4% open a non-gold evidence document, while snippet-only guesses drop from 25.3% at 0-shot to 0.5% and ungrounded parametric guesses drop to 0.0%. At serving time, unlike in-place context edits (CLM+SCR [15], which incur 41.8% edit re-prefill and invalidate $\mathbf{h}_t$), Octopus reuses 89.59% of prompt tokens from the GPU KV cache across 447,993 H200 prefill batches, cuts wire traffic by 98.3% (662.3 $\rightarrow$ 11.5 MB), and accelerates 128K prompt assembly by 249× (358.5 $\rightarrow$ 1.44 ms).

Table 4: Per-episode context conversion efficiency on BrowseComp-Plus (830 queries) and server-resident radix prefix reuse across 447,993 live H200 prefill batches.

Context Management & Serving Architecture	Explored Context (Active / Win. Cap)	Critical-Path Token Load	Gold Open Rec. / BCP Acc. ↑	GPU Prefix KV Cache Reuse ↑	128K Assembly & Wire Cost ↓
Single-Agent ReAct (Stateless 32K)	32.6K (100% of 32K cap)	1,043 tok (1.00×)	12.2% / 26.0%	Append-only ($\mathcal{O}(L)$ resend)	358.5 ms / 662.3 MB
Single-Agent GRPO (Flat 143K Window)	40.8K (28.5% of 143K cap)	4,075 tok (3.9× bloat)	21.1% / 32.1%	Polluted window (0% pruned)	358.5 ms / 662.3 MB
In-Place Context Edit (CLM+SCR, 9B) [15]	Multi-edit ($\leq$ 32K active cap)	1.00× serial edits	— / 42.5% (1.34 PF)	41.8% edit re-prefill (breaks $\mathbf{h}_t$)	Suffix copy & re-rotate
Octopus (Ours, 32K Branches) **Server-Resident Radix Cursor**	**329.2K expl. / 199.0K net** (26.4K active; 541K hard)	**1.94× compressed** (9,919/19,251 tok)	43.3% / 49.6% (58.9% $\geq$ 1 **gold;** +22.2)	**89.59% GPU KV Hit** (0% prefix re-prefill)	**1.44 ms** (249× ↑) (11.5 MB, −98.3%)

Hijack-Free Cross-Branch KV Mounting. Figure 9(b) and Table 5 benchmark cross-branch sharing across 366 live `read_thread` mounts on Qwen3.5-4B (947 exact-match identifier probes and 368 fact-recall probes; in 52% of mounts, sibling A ran longer than reader B, $e_A > p_B$). Copying A's transcript as text (**Transcript Copy** [5]) or splicing KV caches across all layers including recurrent states (**Full-Layer KV Splice** [20]) allocates +100% extra KV

memory and steps B's 24 Gated DeltaNet states $\mathbf{h}_B$ through A's tokens. As predicted in Section 3.2, overwriting $\mathbf{h}_B$ distorts B's policy (ΔNLL $= +0.195$–0.196) and hijacks 55.2%–57.9% of post-read turns into pursuing A's sub-goal. Mounting attention KV without shifting positions (**Unshifted Zero-Copy KV**) avoids memory copies but drops from 0.612 to 0.449 exact match whenever $e_A > p_B$ due to inverted RoPE phases, whereas copying and re-rotating keys (**RoPE-Relocated KV Copy** [21, 22]) doubles KV memory and still incurs 7.1% hijacking.

Octopus Zero-Copy Mount (Rule K2, Equation (2)) splices sibling KV pages exclusively into the 8 full-attention layers under monotonic position offsets while all 24 recurrent layers skip the span: it allocates 0 extra KV pages, retains 92.2% of full-text answer log-probability gain (96.5%–97.3% fact recall) with no degradation on longer siblings (0.613 vs. 0.612 exact match), cuts policy drift by 29× (ΔNLL $= +0.007$), and reduces off-task hijacking by 21× (57.9% → 2.7%). Across 115 live rollouts, a blind judge finds no excess drift relative to a control that blocks the read (ΔHijack $= -1.1$%), and the mechanism transfers cleanly to dense Transformers (Qwen3-4B, 96.9% argmax agreement) and Mamba-2 hybrids (Granite-4.0-H, 98.5%; Appendix C).

Table 5: Cross-branch context-sharing efficiency across 366 live `read_thread` mounts (947 probes on Qwen3.5-4B; 52% with longer sibling $e_A > p_B$).

Cross-Branch Context-Sharing Mechanism	Extra KV Memory ↓	Full-Text Info Retained ↑	Exact Match ↑ All (Longer Sib. $e_A > p_B$)	Policy Drift ΔNLL ↓	Off-Task Hijack Rate ↓
No Sibling Read (`none`)	0 Pages	0.0% (62.5% rec.)	0.031 (0.032)	0.000	0.5%
Transcript Copy (AutoGen [5])	+100% (+L_{mount})	**100%** (**99.5%** rec.)	**0.760** (0.786)	+0.196	57.9%
Full-Layer KV Splice (CacheBlend [20])	+100% (+L_{mount})	98.1% (99.2% rec.)	0.715 (0.731)	+0.195	55.2%
Unshifted Zero-Copy KV	0 Pages (0-Copy)	86.1% (94.3% rec.)	0.521 (0.449)	+0.003	2.5%
RoPE-Relocated KV Copy (KVCOMM [21, 22])	+100% (+L_{mount})	91.7% (96.7% rec.)	0.601 (0.611)	+0.015	7.1%
Octopus Zero-Copy Mount (Ours) **Attn-Only Monotonic Offset (Equation (2))**	**0 Pages (0-Copy)**	92.2% **Info** (96.5%–97.3% rec.)	0.612 (0.613) **(0-shot: 0.627)**	+0.007 (29× ↓ vs. Text)	2.7% (3.8% **0-shot**) **(21× ↓ vs. Text)**

Cross-Architecture Transfer: Qwen3-4B (Dense, 9/36L) **96.9%** · Granite-4.0-H (Mamba-2 Hybrid) **98.5%**. **Live Blind Audit** (115 **Reads**): ΔHijack $= -\mathbf{1.1}$% vs. Blocked-Read Control.

4.3 Multi-Agent Scalability Across Agent Count and Tree Depth

Scaling Accuracy with Deployed Agents and Test-Time Fork Ablation. Figure 10(a) and Table 6 track performance from a single linear thread ($W_{\max} = 0$) to depth-1 parallel teams ($D_{\max} = 1, W_{\max} = 10$) and unlimited-depth hierarchies ($D_{\max} = \infty, W_{\max} = 10$, up to 11 threads/episode). Across 25,812 depth-1 training rollouts, branched trajectories reach 61%–66% accuracy versus 52% for unbranched rollouts on the same query, whereas rewinding an unbranched root yields no gain ($\Delta y = -0.014$); accordingly, DAG-GRPO drives root forking from 27% to 98% while cutting root rewinds 15× (0.413 → 0.026/episode). Held-out accuracy climbs monotonically with deployed agents per episode (Figure 10(a)): on OpenResearcher Validation, accuracy rises from 40.0% (32K ReAct) and 43.2% (143K GRPO) to 51.2% (+8.0 points); on BrowseComp-Plus, accuracy scales from 26.0% (32K ReAct) and 32.1% (143K GRPO) to 42.6% at 4.18 agents/episode ($D_{\max} = 1$) and 49.6% at 7.41 agents/episode ($D_{\max} = \infty$, 6.41 workers). Recursive branching rescues 31.8% of queries that hit the 32K single-agent context wall (0% rescued at 0-shot) and boosts high-hop third-quartile (Q3) queries by 10× (3.4% → 34.1%).

Crucially, we cap fanout at $W_{\max} = 10$ only to match 9B single-agent RL compute (2.00 vs. 2.19 PF/Q): episodes finishing naturally within 10 workers (69.2% of queries) achieve 71.8% accuracy while 39.0% hit the 10-worker cap, indicating that scaling is budget-limited rather than saturated. Conversely, disabling `fork` at test time on the exact same Step 45 weights ($W_{\max} = 0$) drops paired BrowseComp-Plus accuracy by 21.7 points (50.8% → 29.1% across 571 paired queries; 45.4% on Validation), confirming that gains come from live multi-agent execution rather than single-thread weight memorization.

Step-15 Phase Transition, Emergent Peer KV Mesh, and 1D Training Scaling. Without our loss-masked termination prompt when a branch omits `done` (Appendix A), naive $D_{\max} = \infty$ RL collapses from 28.2% to 0% forking within 15 steps (Figure 10(b)). With this termination guard (24,041 episodes over 45 steps), Octopus plateaus for 14 steps and then undergoes a sharp topological phase transition at Step 15: deep trees reaching $d \geq 3/4/5$ surge to 59.3%/39.1%/21.5% (54× increase at $d \geq 5$), yet the mean children spawned per worker remains sub-critical at $\mu \in [0.32, 0.69] < 1.0$, bounding expected tree size without depth caps (6.41 workers/episode on BrowseComp-Plus).

Driven solely by binary 0/1 terminal rewards (Figure 10(c,d)), Step 15 simultaneously self-organizes three division-of-labor behaviors: *(i)* Turn-1 root delegation jumps 38× (2.6% → 98.0%) while workers absorb 92.8% of 8K page `open` calls, keeping root contexts clean; *(ii)* branches form a lateral `read_thread` KV mesh (17.3% → 85.9% of

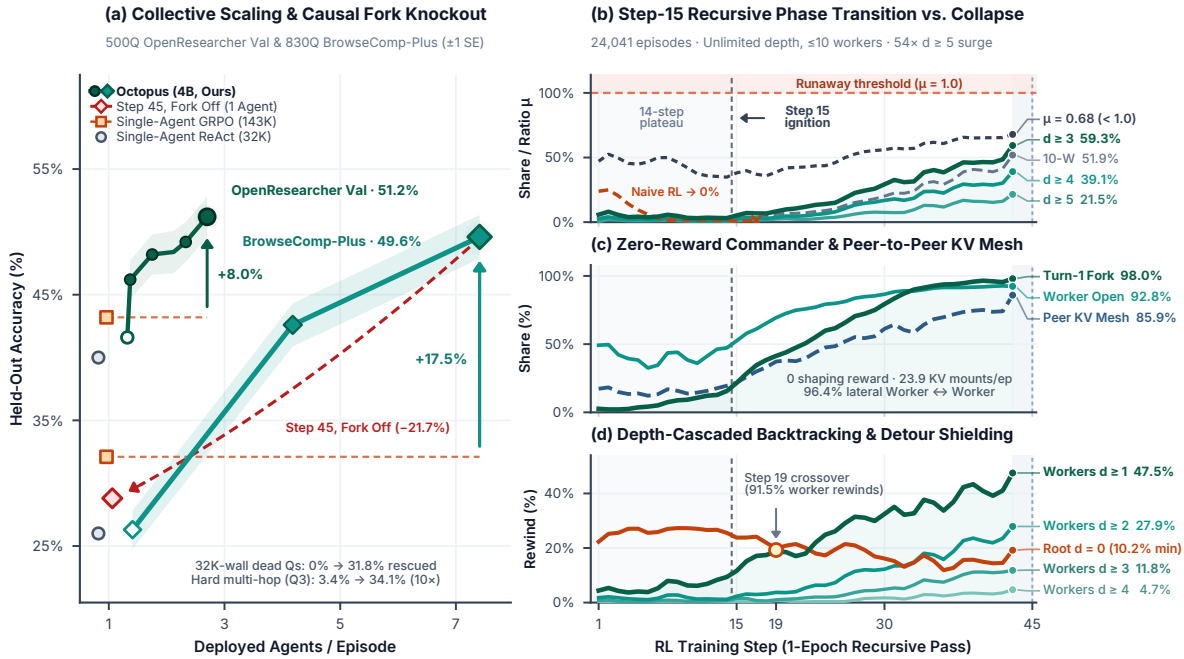


Figure 10: Multi-agent scaling and Step-15 recursive phase transition on Qwen3.5-4B. **(a)** Held-out accuracy vs. deployed agents per episode and same-weight test-time `fork` ablation ($W_{\max} = 0, \pm 1$ SE). **(b–d)** Unlimited-depth RL dynamics (24,041 episodes, $D_{\max} = \infty, W_{\max} = 10$), showing sub-critical recursive scaling ($\mu < 1$), emergent peer-to-peer KV mounting, and depth-cascaded worker `rewind`s.

episodes, averaging 23.9 zero-copy mounts/episode with 96.4% peer worker↔worker reads); and *(iii)* past Step 19, root rewinds drop nearly 3× (29.3% → 10.2% of episodes) as workers absorb 91.5% of all `rewind` calls (47.5% of episodes at $d \geq 1$ and 27.9% at $d \geq 2$; Appendix B). Finally, prefix-tree packed 1D deduplication with live mounted KV edges (Sections 3.3 and 3.4) scales owned generated tokens trained per step by 8.1× on average (11.1× peak, 101.6K → 1.12M tokens/step, 0 OOM) while holding rollout–training divergence at $\mathrm{KL}(\pi_{\text{rollout}} \,\|\, \pi_\theta) = 4.45 \times 10^{-4}$—32× lower than dropping mounted KV edges (0.0142).

Table 6: Multi-agent scalability across worker budget ($W_{\max}$), recursive tree depth ($D_{\max}$), and emergent division of labor on Qwen3.5-4B.

Multi-Agent Topology & Training Regime	Max Threads (Depth, Fanout)	Root Fork Rate	Workers / Ep (Child Ratio μ)	Deep ($d \geq 3$) / Full 10-Br.	Rewind Activity (Root / Worker)	OpenResearcher Acc. ↑	BCP Test Acc. ↑
Single-Agent ReAct (32K Window)	1 (0, 0)	0%	0.00 (0.00)	0% / 0%	— / —	40.0% (Val)	26.0%
Single-Agent GRPO (143K Flat)	1 (0, 0)	0%	0.00 (0.00)	0% / 0%	— / —	43.2% (Val)	32.1%
Multi-Agent Fork (0-Shot Base)	≤ 11 (∞, 10)	11% (16% Val)	0.41 (0.32 Val)	0% / 0%	1.51 rw/ep (9.7 st)	41.6% (Val)	26.3% (24.8%)
Octopus RL (Parallel Team, $D_{\max}$=1, Step 35)	≤ 11 (1, 10)	92%	3.18 (0.00)	0% / 0.2%	0.29/ep / 0.52/br	46.5% **(Val)**	42.6% **(+10.5)**
Octopus RL (Parallel Team, $D_{\max}$=1, Step 55)	≤ 11 (1, 10)	98%	1.70 (0.00)	0% / 0.6%	0.026**/ep** / 0.24**/br**	**51.2% (Val)**	—
Recursive 0-Shot ($D_{\max}$=∞ on $D_{\max}$=1 Policy)	≤ 11 (∞, 10)	97%	2.34 (0.27)	8.6% / —	— / —	44.4% (57.2%‡)	—
Octopus RL (Recursive, $D_{\max}$=∞, Step 20)	≤ 11 (∞, 10)	55%	1.15 (0.46)	11.4% / 8.9%	23.7% / 18.6% ep	46.6% **(Val)**	—
Octopus RL (Recursive, $D_{\max}$=∞, Step 43 Rollout‡)	≤ 11 (∞, 10)	99.2%	6.85 (0.68)	59.3% / 51.9%	10.2% / 47.5% ep	57.9% **(Train‡)**	—
Octopus RL (Recursive, $D_{\max}$=∞, Step 45, 1-Ep.)	≤ 11 (∞, 10)	99.5%	6.41 (7.41 **ag.**)	— / 39.0%	6.99 **rw/ep** (61.4 st)	45.4→46.5% ($W \leq 1$‡)	**49.6% (+17.5)**

Prefix-Tree Packed Deduplication (Section 3.3): **8.1**× avg (**11.1**× peak, 101.6K→1.12M tok/step, 0 OOM). **On-Policy Mount KL:** $\mathbf{4.45 \times 10^{-4}}$ (vs. 0.0142 w/o mounts, **32**× ↓).

‡Zero-shot $D_{\max} = \infty$ transfer of the $D_{\max} = 1$ policy scores 44.4% across all 500 Val queries and 57.2% on completed trees. Step 43 reports late-training rollout statistics (57.9% training rollout accuracy); at Step 45 (1 epoch = 3,040 queries), restricting test-time fanout on Val to $W_{\max} = 0/1$ yields 45.4%/46.5%, while full recursive evaluation ($W_{\max} = 10$) reaches **49.6**% on BrowseComp-Plus.

5 Conclusion and Limitations

Long-horizon reasoning agents need not be confined to a single append-only text window. **Octopus** recasts agentic execution as a stateful, GPU-resident DAG in which branches fork and backtrack over reference-counted radix KV prefixes and recurrent checkpoints in $\mathcal{O}(1)$ zero copy, while exchanging findings directly through attention-only KV mounting without overwriting private recurrent working memory (92.2% information retention, $21\times$ lower task hijacking, 89.6% live GPU KV hit rate). Trained end-to-end with prefix-tree packed DAG-GRPO under pure binary outcome rewards and zero process shaping, a 4B sparse-BM25 model self-organizes a sub-critical commander–worker hierarchy and peer KV mesh—reaching 49.6% on BrowseComp-Plus ($+17.5$ points over 143K single-agent RL; -21.7 points when test-time `fork` is disabled on the same weights) and 51.2% on OpenResearcher, outperforming dense-retriever systems up to $250\times$ larger. These results establish latent-space multi-agent branching as an effective scaling axis orthogonal to model parameters and single-window length.

Limitations and Future Directions. Our current study trains a single homogeneous backbone over episode-scoped KV cursors within single-node GPU pools (up to 524 resident cursors/GPU at $W_{\max} = 10$, with tree-size loss normalization $1/Z_i$), opening three broader directions. First, **heterogeneous multi-model branching**: while our runtime already supports seven dense, MoE, and hybrid families (Appendix C), each episode currently executes one shared model; routing sub-branches across role-specialized LoRA adapters or heterogeneous model scales—such as lightweight scout branches paired with heavyweight verifiers via learned KV projections—could markedly improve compute–accuracy trade-offs. Second, **persistent LoRA and latent long-term memory**: mounted KV pages are currently reclaimed at episode end; consolidating verified branch KV states into persistent cross-episode memory banks or continual LoRA weights would extend Octopus from episodic reasoning to lifelong agentic memory. Third, **million-agent collective scaling**: tiering reference-counted radix cursors and recurrent checkpoints across GPU HBM, host DRAM/NVMe, and multi-node RDMA fabrics would scale latent-space coordination from ten-branch trees to asynchronous million-agent swarms.

References

[1] Bowen Jin, Hansi Zeng, Zhenrui Yue, Jinsung Yoon, Sercan Arik, Dong Wang, Hamed Zamani, and Jiawei Han. Search-R1: Training LLMs to reason and leverage search engines with reinforcement learning. In *Conference on Language Modeling (COLM)*, 2025.

[2] Zijian Chen, Xueguang Ma, Shengyao Zhuang, Ping Nie, Kai Zou, Andrew Liu, Joshua Green, Kshama Patel, Ruoxi Meng, Mingyi Su, et al. BrowseComp-Plus: A fair and disentangled evaluation benchmark for deep search agents. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026.

[3] Zhuofeng Li, Dongfu Jiang, Xueguang Ma, Haoxiang Zhang, Ping Nie, Yuyu Zhang, Kai Zou, Jianwen Xie, Yu Zhang, and Wenhu Chen. OpenResearcher: A fully open pipeline for long-horizon deep research trajectory synthesis. *arXiv preprint arXiv:2603.20278*, 2026.

[4] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik R. Narasimhan, and Yuan Cao. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations (ICLR)*, 2023.

[5] Qingyun Wu, Gagan Bansal, Jieyu Zhang, Yiran Wu, Beibin Li, Erkang Zhu, Li Jiang, Xiaoyun Zhang, Shaokun Zhang, Jiale Liu, et al. AutoGen: Enabling next-gen LLM applications via multi-agent conversation. In *Conference on Language Modeling (COLM)*, 2024.

[6] Sirui Hong, Mingchen Zhuge, Jonathan Chen, Xiawu Zheng, Yuheng Cheng, Jinlin Wang, Ceyao Zhang, Zili Wang, Steven Ka Shing Yau, Zijuan Lin, et al. MetaGPT: Meta programming for a multi-agent collaborative framework. In *International Conference on Learning Representations (ICLR)*, 2024.

[7] Zihan Wang, Kangrui Wang, Qineng Wang, Pingyue Zhang, Linjie Li, Zhengyuan Yang, Xing Jin, Kefan Yu, Minh Nhat Nguyen, Licheng Liu, et al. RAGEN: Understanding self-evolution in LLM agents via multi-turn reinforcement learning. In *International Conference on Machine Learning (ICML)*, 2026.

[8] Wei Fu, Jiaxuan Gao, Xujie Shen, Chen Zhu, Zhiyu Mei, Chuyi He, Shusheng Xu, Guo Wei, Jun Mei, Jiashu Wang, et al. AReaL: A large-scale asynchronous reinforcement learning system for language reasoning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

[9] Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics (TACL)*, 12:157–173, 2024.

[10] Tianyu Fu, Ruoxi Wang, Zihan Min, Zhongqi Ren, Zhihao Li, Tengxuan Liu, Shengxu Mu, Hanling Zhang, and Yu Wang. Cache engineering for agents. *arXiv preprint arXiv:2610.28577*, 2026.

[11] Federico Cassano and Sasha Rush. Training Composer for longer horizons. Cursor Research Blog, `https://cursor.com/blog/self-summarization`, 2026.

[12] Xinyu Wu et al. ReSum: Unlocking long-horizon search intelligence via context summarization. *arXiv preprint arXiv:2509.13313*, 2025.

[13] Zijian Zhou, Ao Qu, Zhaoxuan Wu, Sunghwan Kim, Alok Prakash, Daniela Rus, Jinhua Zhao, Bryan Kian Hsiang Low, and Paul Pu Liang. MEM1: Learning to synergize memory and reasoning for efficient long-horizon agents. In *International Conference on Learning Representations (ICLR)*, pages 58413–58438, 2026.

[14] Xiaochuan Li, Ryan Ming, Meng Chu, Shuai Shao, Rong Jin, and Chenyan Xiong. ACM: Agentic context management for long horizon tasks. *arXiv preprint arXiv:2607.23809*, 2026.

[15] Rulin Shao, Shannon Zejiang Shen, Junjie Oscar Yin, Yuetai Li, Minheng Wang, Hamish Ivison, Radha Poovendran, Nathan Lambert, Teng Xiao, Mike Lewis, Wen-tau Yih, Luke Zettlemoyer, and Pang Wei Koh. Context language models. *arXiv preprint arXiv:2609.37725*, 2026.

[16] Songlin Yang, Jan Kautz, and Ali Hatamizadeh. Gated delta networks: Improving Mamba2 with delta rule. In *International Conference on Learning Representations (ICLR)*, 2025.

[17] Tri Dao and Albert Gu. Transformers are SSMs: Generalized models and efficient algorithms through structured state space duality. In *International Conference on Machine Learning (ICML)*, 2024.

[18] Guohao Li, Hasan Abed Al Kader Hammoud, Hani Itani, Dmitrii Khizbullin, and Bernard Ghanem. CAMEL: Communicative agents for "mind" exploration of large language model society. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[19] Arpandeep Khatua, Hao Zhu, Peter Tran, Arya Prabhudesai, Frederic Sadrieh, Johann K. Lieberwirth, Xinkai Yu, Yicheng Fu, Michael J. Ryan, Jiaxin Pei, and Diyi Yang. CooperBench: Why coding agents cannot be your teammates yet. In *International Conference on Learning Representations (ICLR)*, 2026.

[20] Jiayi Yao, Hanchen Li, Yuhan Liu, Siddhant Ray, Yihua Cheng, Qizheng Zhang, Kuntai Du, Shan Lu, and Junchen Jiang. CacheBlend: Fast large language model serving for RAG with cached knowledge fusion. In *Proceedings of the Twentieth European Conference on Computer Systems (EuroSys)*, pages 94–109, 2025.

[21] Gleb Rodionov, Roman Garipov, Alina Shutova, George Yakushev, Erik Schultheis, Vage Egiazarian, Anton Sinitsin, Denis Kuznedelev, and Dan Alistarh. Hogwild! Inference: Parallel LLM generation via concurrent attention. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

[22] Hancheng Ye, Zhengqi Gao, Mingyuan Ma, Qinsi Wang, Yuzhe Fu, Ming-Yu Chung, Yueqian Lin, Zhijian Liu, Jianyi Zhang, Danyang Zhuo, and Yiran Chen. KVCOMM: Online cross-context KV-cache communication for efficient LLM-based multi-agent systems. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

[23] Hyesung Jeon, Hyeongju Ha, Seoyoung Lee, Beomseok Kang, and Jae-Joon Kim. KVCMAS: Efficient KV cache correction for shared context in multi-agent systems. *arXiv preprint arXiv:2609.34060*, 2026.

[24] Jiayi Pan, Xiuyu Li, Long Lian, Charlie Snell, Yifei Zhou, Adam Yala, Trevor Darrell, Kurt Keutzer, and Alane Suhr. Learning adaptive parallel reasoning with language models. *arXiv preprint arXiv:2504.15466*, 2025.

[25] Weiwei Sun, Miao Lu, Zhan Ling, Kang Liu, Xuesong Yao, Yiming Yang, and Jiecao Chen. Scaling long-horizon LLM agent via Context-Folding. In *International Conference on Machine Learning (ICML)*, 2026.

[26] Haoyang Hong, Jiajun Yin, Yuan Wang, Jingnan Liu, Zhe Chen, Ailing Yu, Ji Li, Zhiling Ye, Hansong Xiao, Yefei Chen, et al. Multi-agent deep research: Training multi-agent systems with M-GRPO. *arXiv preprint arXiv:2511.13288*, 2025.

[27] Kirill Afendulev, Alexey Dontsov, Elena Tutubalina, and Anton Korznikov. What attention recalls and recurrence controls in hybrid language models. *arXiv preprint arXiv:2609.04434*, 2026.

[28] DeepSeek-AI, Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, et al. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *Nature*, 645, 2025.

[29] Song-Li Wu, Jingyi Wang, Zhaocheng Du, and Weinan Gan. ActiveMem: Dynamic latent memory trees for long-horizon agents. *arXiv preprint arXiv:2609.33244*, 2026.

[30] Tianjian Li, Jingyu Zhang, William Jurayj, Xi Wang, Chuanyang Jin, Mehrdad Farajtabar, Eric Nalisnick, and Daniel Khashabi. Self-compacting language model agents. *arXiv preprint arXiv:2606.23525*, 2026.

[31] Mo Li, L. H. Xu, Qitai Tan, Long Ma, Hongyong Song, Ting Cao, and Yunxin Liu. Sculptor: Empowering LLMs with cognitive agency via active context management. In *International Conference on Learning Representations (ICLR)*, pages 153411–153440, 2026.

[32] Alex L. Zhang, Tim Kraska, and Omar Khattab. Recursive language models. *arXiv preprint arXiv:2512.24601*, 2025.

[33] Vasilis Kontonis, Yuchen Zeng, Shivam Garg, Lingjiao Chen, Hao Tang, Ziyan Wang, Ahmed Awadallah, Eric Horvitz, John Langford, and Dimitris Papailiopoulos. Memento: Teaching LLMs to manage their own context. *arXiv preprint arXiv:2604.09852*, 2026.

[34] Xiangyu Shi, Marco Chiesa, Gerald Q. Maguire Jr., and Dejan Kostić. KVComm: Enabling efficient LLM communication through selective KV sharing. In *International Conference on Learning Representations (ICLR)*, 2026.

[35] In Gim, Guojun Chen, Seung-seob Lee, Nikhil Sarda, Anurag Khandelwal, and Lin Zhong. Prompt cache: Modular attention reuse for low-latency inference. In *Proceedings of Machine Learning and Systems (MLSys)*, 2024.

[36] Junhao Hu, Wenrui Huang, Weidong Wang, Haoyi Wang, Tiancheng Hu, Qin Zhang, Hao Feng, Xusheng Chen, Yizhou Shan, and Tao Xie. EPIC: Efficient position-independent caching for serving large language models. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, pages 24391–24402, 2025.

[37] Yuhan Liu, Hanchen Li, Yihua Cheng, Siddhant Ray, Yuyang Huang, Qizheng Zhang, Kuntai Du, Jiayi Yao, Shan Lu, Ganesh Ananthanarayanan, et al. DroidSpeak: KV cache sharing for cross-LLM communication and multi-LLM serving. In *USENIX Symposium on Networked Systems Design and Implementation (NSDI)*, 2026.

[38] Tianyu Fu, Zihan Min, Hanling Zhang, Jichao Yan, Guohao Dai, Wanli Ouyang, and Yu Wang. Cache-to-cache: Direct semantic communication between large language models. In *International Conference on Learning Representations (ICLR)*, 2026.

[39] Ye Yu, Heming Liu, Haibo Jin, Xiaopeng Yuan, Peng Kuang, and Haohan Wang. Learning to communicate: Toward end-to-end optimization of multi-agent language systems. *arXiv preprint arXiv:2604.21794*, 2026.

[40] Jiarui Zhang, Yuchen Yang, Ran Yan, Zhiyu Mei, Liyuan Zhang, Daifeng Li, Wei Fu, Jiaxuan Gao, Shusheng Xu, Yi Wu, and Binhang Yuan. AREAL-DTA: Dynamic tree attention for efficient reinforcement learning of large language models. *arXiv preprint arXiv:2602.00482*, 2026.

[41] Weiqi Wang, Yuxin Zhou, Mouxiang Chen, Siyuan Zhang, Yi Zhang, Yuyan Luo, Zhiyu Yin, Chencan Wu, Jiemin Jiang, Wentao Yao, Chujie Zheng, and JianWei Zhang. QwenGyre: An elastic reinforcement learning framework for training xLong-horizon agents. *arXiv preprint arXiv:2609.33848*, 2026.

[42] Yuhan Sun, Jiaxin Zhao, Yongdong Luo, Yibo Hu, Meiguang Jin, Junfeng Ma, Weixun Wang, Weihang Pan, and Zhongan Bi. MultiAgent-OPD: End-to-end training of hierarchical multi-agent systems with on-policy distillation. *arXiv preprint arXiv:2610.23234*, 2026.

[43] Wouter Kool, Herke van Hoof, and Max Welling. Buy 4 REINFORCE samples, get a baseline for free! In *ICLR Workshop on Deep Reinforcement Learning Meets Structured Prediction*, 2019.

[44] Zichen Liu, Changyu Chen, Wenjun Li, Penghui Qi, Tianyu Pang, Chao Du, Wee Sun Lee, and Min Lin. Understanding R1-Zero-Like training: A critical perspective. *arXiv preprint arXiv:2503.20783*, 2025.

A Cross-Family Chat-Template Probing and Runtime Guardrails

Automatic Startup Chat-Template Probing and Tagged Injections. Open-weight chat templates $\mathcal{T}$ frequently violate strict append-only rendering: Qwen2.5/3/3.5 strip earlier assistant reasoning blocks upon a new user turn, DeepSeek-V3/R1 format initial and subsequent tool calls asymmetrically, Mistral-v0.3 embeds the system prompt inside the final user turn, and gpt-oss drops analysis channels on re-render. At server startup (~ 0.1 s), Octopus probes synthetic $(\text{parent-reply}, \text{delta-turn})$ pairs to locate the assistant-turn terminator τ_{term} and classify $\mathcal{T}$ into one of two incremental delta modes (Section 3.2). Templates with history-invariant suffixes slice Render_{Δ} directly after τ_{term} (covering Qwen, Llama-3, DeepSeek-V3.1, and GLM-4.5/4.7), whereas context-sensitive templates render against a minimal prefix $\text{ctx}(c)$ before slicing (covering DeepSeek-V3/R1, Mistral-v0.3, and gpt-oss), yielding bit-exact tokenization across all 18,344 test turns. Mid-episode runtime notifications (`fork`, join, and `rewind` notes) are injected as tagged tool responses rather than user turns so prior reasoning blocks remain intact.

Dynamic `read_thread` Schema Injection, Answer Reserve, and Eviction Rebuild. To preserve 100% shared prefix KV caching while preventing pre-fork self-reads, Octopus omits `read_thread` from the static system prompt and appends its tool schema and sibling roster only after a peer step completes, clamping requested step ranges $[s, e]$ to the source's valid steps $[A, B]$ via $[\max(s, A), \min(e, B)]$. Within each thread's $B_{\text{thread}} = 32{,}768$-token growth budget, $R_{\text{ans}} = 2{,}048$ tokens are reserved for final synthesis, and plain-text replies without a tool call trigger a single loss-masked termination prompt ($m_{\ell,t} = 0$). Completed cursors record step-boundary offsets in a server-side LRU registry, while mounted KV spans link into the GPU radix tree through reference-counted page pointers. If GPU memory pressure or an online LoRA weight update evicts a mounted segment, the server transparently re-prefills the span from the cursor's step offsets before decoding.

KV-Aware Episode Admission and Branch Turn Budget. Gating new episode admission on live server KV pool utilization ($\tau_{\text{KV}} = 0.75$ with zero waiting queue) prevents mid-episode fork bursts from saturating GPU memory (Section 4.3). Whereas unbounded dispatch drives P90 KV utilization to 98%–99% (72–88 queued requests) and causes 14.7%–46.1% episode timeouts, the 75% gate caps P90 KV load at 74.5% with zero queued requests and zero timeouts across 1,304 episodes. This allows each 141GB GPU (1.94M-token KV pool) to host up to 524 resident GDN recurrent checkpoints and 154 concurrently decoding threads at 1,044.5 tok/s (P90). In addition, raising the per-branch turn cap from $T_{\text{br}} = 20$ to $T_{\text{br}} = 50$ prevents rewound branches from exhausting their turn budget before calling `done`, narrowing the completion-rate gap between rewound and unrewound branches (ΔDone) from -36.0% to -4.1%.

Termination Prompt Guardrail Against Policy Collapse. We further compare three episode-termination regimes across 33,075 pooled unlimited-depth rollouts ($D_{\max} = \infty, W_{\max} = 10$). Under *fallback plain-text grading*, scoring tool-less root replies via answer extraction causes unverified tool-less endings to surge from 6% to 75% by Step 12 (41% at $d \geq 3$). Conversely, under *strict `done`-only grading without a termination prompt*, the 7.5% of forked roots that emit plain text after a join note receive zero reward ($y = 0$), assigning negative advantage to forking ($\hat{A}_{\text{fork}} = -0.044$) and collapsing root fork rates from 23.9%–28.2% to 0% at Step 15 (Figure 10(b)). Injecting our single-turn loss-masked termination prompt ($m_{\ell,t} = 0$) converts 97.4% of reminded roots and 93.1% of reminded sub-branches into valid `done` calls (0.2% tool-less), sustaining 98.8%–99.2% root forking across the full 45-step training pass.

B Concrete Unlimited-Depth Search Rollout Walkthrough (`search-3190-99`)

To illustrate how Octopus's five execution primitives, layer-selective KV mounting, and sub-critical unlimited-depth tree execution work together on a long-horizon episode, Figure 11 and Table 7 trace a real evaluation episode (`search-3190-99`) on the multi-hop query: *"During a late twentieth-century volcanic event in the Luzon Volcanic Arc that was notable for lowering global temperatures for several years and having an ash cloud significantly larger than the 1980 eruption of a U.S. mountain, how many commercial airplanes experienced in-flight damage as a result of inadvertently entering the airborne particulate cloud generated by this eruption? Answer with a whole number only."*

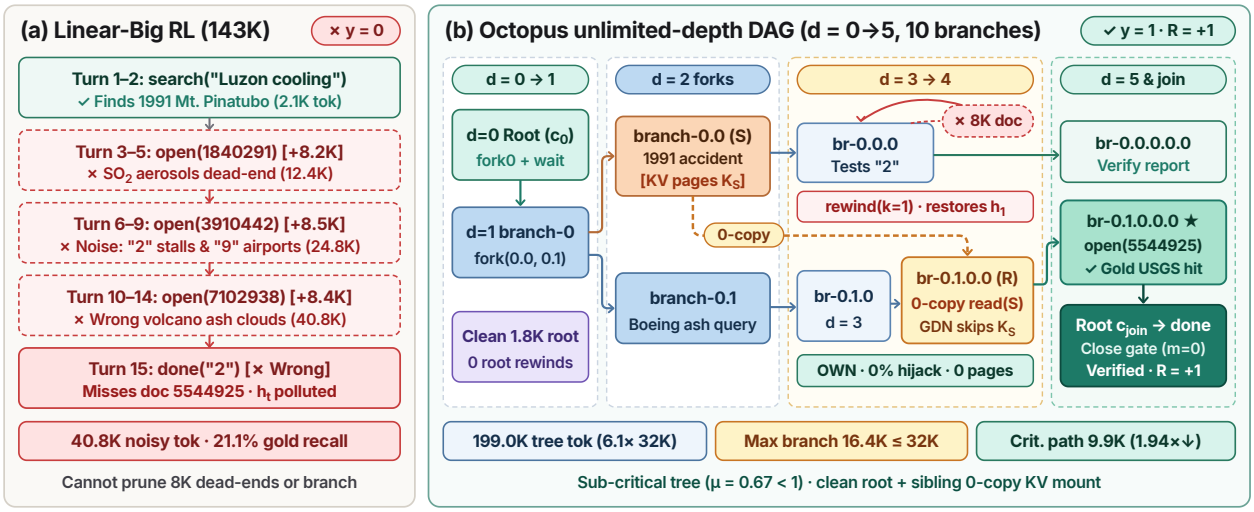


Figure 11: Multi-hop search rollout case study (`search-3190-99`: 1991 Mt. Pinatubo ash-cloud aircraft damage). **(a)** Flat 143K single-thread RL (32.1% BCP) accumulates noisy retrieval detours inside one append-only context, polluting its recurrent state $\mathbf{h}_t$. **(b)** Under $D_{\max} = \infty, W_{\max} = 10$, Octopus (49.6% BCP, 57.9% Rollout) keeps the root clean (1.8K tok) while spawning a depth-5, 10-branch DAG ($\mu = 0.67 < 1$): dead-end queries are pruned via branch `rewind`, and `branch-0.1.0.0` ($d = 4$) zero-copy mounts sibling `branch-0.0`'s attention KV pages (0% hijack) before opening `docid=5544925` at $d = 5$.

Table 7: Step-by-step trace and hybrid recurrent/attention state transitions for unlimited-depth rollout `search-3190-99` ($D_{\max} = \infty, W_{\max} = 10$, 10 branches across $d = 0 \to 5$).

Depth & Cursor	Agent Trajectory (Prompt / Thought / Tool Call & Return)	GPU Radix Cursor, Hybrid State & DAG-GRPO Semantics
$d = 0$ **Root** $c_0 \to$ `fork0` (Orchestrator)	**Thought:** Identify 1991 Mt. Pinatubo (Luzon Arc) and delegate commercial aircraft damage lookup. **Call:** `fork({"branch-0": "find number of commercial airplanes that experienced in-flight damage from Mount Pinatubo 1991 ash cloud"}) + wait(["branch-0"])`	✓ $\mathcal{O}(1)$ **Zero-Copy Spawn:** Increments Radix KV ref-count at c_{fork} by $+1$; clones GDN recurrent state $\mathbf{h}_{\text{end}}(c_0)$. ✓ **Clean Orchestrator:** Root suspends at 1.8K tok; `branch-0` gets fresh $B_{\text{thread}} = 32$K cap.
$d = 1 \to 2$ **Fork** `branch-0` $\to$ `0.0` (S), `0.1`	**Call** ($d = 1$)**:** `search("Mount Pinatubo 1991 ash cloud commercial airplanes damage")` $\to$ surfaces "more than 20 airplanes" and "24 windshields", then sub-forks ($d = 2$): • `branch-0.0` (S): `"search for Pinatubo 1991 aircraft accident airplanes"` • `branch-0.1`: `"search for \"Pinatubo\" Boeing airplanes ash"`	✓ **Recursive Expansion** ($D_{\max} = \infty$)**:** Each $d = 2$ child shares parent KV prefix in $\mathcal{O}(1)$ and inherits $\mathbf{h}_{\text{end}}$(`branch-0`). ✓ **Hypothesis Isolation:** Sibling S (`branch-0.0`) extracts candidate docID `5544925` and encounter tables into its KV pages.
$d = 3 \to 5$ **Left** `branch-0.0.0*` (Dead-End Path)	**Sub-Forks:** `0.0.0` ($d = 3$: `"...damage 2"`) $\to$ `0.0.0.0` ($d = 4$) $\to$ `0.0.0.0.0` ($d = 5$). **Dead End & Rewind:** Queries on "2 crashes" open noisy incident pages (+8.4K tok); branch calls `rewind(step=1, note="2 is engine replacements; pivot to USGS report")`.	✓ **Branch Rewind:** Cursor resets to c_1 (reclaiming +8.4K of B_{thread}); GDN layers restore exact $\mathbf{h}_1$ (erasing dead-end noise). ✓ **Prefix-Deduplicating Mask (Section 3.4):** Dead-end leaf kept in DAG so `rewind` gets RL credit; post-rewind masks $m_{\ell,t} = 0$ ($t \le 1$).
$d = 3 \to 4$ **Right** `branch-0.1.0.0` (Reader R)	**Sub-Forks:** `branch-0.1.0` ($d = 3$: `"...\"airplanes\" 24 windshields"`) spawns $d = 4$ siblings `0.1.0.0` (R) and `0.1.0.1`. **Cross-Branch Mount:** Reader R (`branch-0.1.0.0`) invokes `read_thread(branch="branch-0.0", steps=[1, 3])` to inspect sibling S's retrieval table without re-searching.	✓ **Hijack-Free Zero-Copy KV Mount (Equation (2)):** • **Full-Attention Layers:** Borrow S's physical KV pages (0 new pages); R's next token jumps to $s(c'_R) = \max(p_R, e_S + 1)$. • **GDN Recurrent Layers:** Skip stepping over mounted span; $\mathbf{h}_R$ advances only along R's post-mount tokens (**retains own sub-task,** 0% **hijack**).
$d = 5$ **Leaf & Join** `0.1.0.0.0` $\to c_{\text{join}}$ (Verify & Close)	**Leaf Verification** ($d = 5$)**:** R spawns `branch-0.1.0.0.0` (`"open 5544925 offset 0"`), verifying the primary USGS/ICAO ash report, and closes via `done`. **Root Synthesis** ($d = 0$)**:** `wait` returns at c_{join}; runtime injects loss-masked termination prompt ($m_{\ell,t} = 0$), and root calls `done` with exact whole-number answer.	✓ **Sub-Critical Tree Growth (Section 4.3):** Terminates at $W = 10$ branches $(1, 2, 2, 3, 2$ at $d = 1..5$; empirical $\mu = 0.67 < 1)$. ✓ **On-Policy Mount Retention (Equations (4) and (8)):** Keeps mounted $\mathbf{K}_S, \mathbf{V}_S$ in the prefix-tree packed forward–backward graph.

C Per-Architecture Framework Integration Blueprints

Octopus supports dense, sparse-MoE, and hybrid recurrent-attention model families across both serving and prefix-tree packed RL through a declarative architecture registry paired with automatic LoRA target resolution (Table 8).

Pure Full-Attention Dense and Sparse-MoE Transformers (`Qwen3 Dense`, `Llama-3`, `Qwen3 MoE`). In models with only full, sliding, or chunked attention and no recurrent state ($\mathbf{h}_t = \emptyset$), `fork` and `rewind` reduce to $\mathcal{O}(1)$ PagedAttention radix pointer updates. Because pure-attention models lack private recurrent layers to anchor branch identity, mounting sibling KV pages across all L layers can induce cross-branch task drift; Octopus therefore mounts $c_A[s:e)$ under monotonic position offsets (Equation (2)) into only a sparse subset of middle retrieval layers ($\mathcal{L}_{\text{mount}} \subset \{1, \ldots, L\}$, e.g., $9/36$ layers in Qwen3-4B-Base) while presenting a mount-free KV view to the remaining 27 layers. This middle-layer mounting retains 96.9% argmax agreement with full-context replay during serving, while the prefix-tree packed trainer dispatches separate variable-length FlashAttention passes for mounted and unmounted layer groups (Table 8). For sparse MoE models (`Qwen3 MoE`), token-wise Top-K routing ($K = 8$) carries no cross-position state, so branching incurs zero expert-state copies and prefix-tree deduplication batches active branches into one grouped GEMM pass.

Hybrid Recurrent-Attention Architectures (`Qwen3.5 GDN` and `Granite-4.0-H Mamba-2`; Figure 12). In Qwen3.5 (8 blocks of three Gated DeltaNet layers and one Gated Full-Attention layer; Figure 12(a)), all 8 full-attention layers zero-copy mount sibling KV pages under Equation (2) (Rule K2 in Figure 12) prior to the output sigmoid gate, whereas all 24 GDN layers clone or restore $\mathbf{h}_k$ on `fork`/`rewind` and skip mounted spans. During prefix-tree packed DAG-GRPO training, tree-boundary hooks gather $d_{\text{conv}} - 1 = 3$ ancestor tokens for short causal convolutions and run level-by-level variable-length GDN recurrence via FlashLinearAttention (`fla`) initialized from each node's parent checkpoint $\mathbf{h}_{\text{end}}(p(u))$. Activation tiling in 16K-token chunks prevents backward memory spikes on trees up to 262K tokens, while fused projection LoRA adapters (Table 8) align PEFT gradients with SGLang serving weights. Analogously, for Mamba-2 + GQA hybrids (`Granite-4.0-H`; Figure 12(b)), Octopus maps Mamba-2 SSD onto `fla`'s chunked linear-attention kernels with ancestor convolution gathering and supports recursive transitive KV mounting ($X \rightsquigarrow A \rightsquigarrow C$, 98.5% token agreement).

Table 8: Per-architecture integration hooks across dense, MoE, and hybrid model families in Octopus.

Architecture Family	Decoder Layer Topology	Serving KV Mount & State Hook	Prefix-Tree Recurrence, Conv & Tiling Hook	Resolved SGLang & Trainer LoRA Targets
`Qwen3 (Dense)` (`qwen3`, `llama3`)	L Full-Attn + SwiGLU MLP ($\mathcal{L}_{\text{rec}} = \emptyset$)	Sparse middle-layer mount ($9/36$ layers mount $c_A[s:e]$); 27 layers unmounted)	No recurrent state; dual variable-length FlashAttn pass (mounted vs. unmounted layers, 32K chunks)	7 modules: `q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj`
`Qwen3 (MoE)` (`qwen3_moe`)	L Full-Attn + Top-K Sparse MoE ($K = 8$)	Attention-only Radix KV mount + Equation (2) jump; 0 MoE state on `fork/rewind`	No recurrent state; deduplicated tree tokens T_{tree} batched into single MoE grouped GEMM	Attention LoRA (`q/k/v/o_proj`); preserves Top-K router gate stability
`Qwen3.5 (Dense)` (`qwen3_5`, Primary)	$8 \times [3$ GDN$+1$ Gated Attn$]$ (24 GDN, 8 Attn) + SwiGLU MLP	All 8 Full-Attn layers 0-copy mount $c_A[s:e)$; 24 GDN layers clone/restore $\mathbf{h}_k$ & skip mounts	$d_{\text{conv}} - 1 = 3$ ancestor conv tokens; chunked GDN recurrence (`fla`) with 16K activation tiling	10 modules: `q/k/v/o_proj, gate/up/down_proj, out_proj` + fused `in_proj_qkvz`
`Qwen3.5-MoE / Next` (`qwen3_5_moe`, `qwen3_next`)	24 GDN + 8 Gated Attn + Sparse Top-K MoE (+ Shared Expert)	8 Full-Attn layers mount $c_A[s:e)$; 24 GDN layers skip mounted spans; 0 MoE state overhead	3-token ancestor conv + level-by-level GDN tree recurrence (32K segments) + grouped MoE forward	Attention + GDN projections (`q/k/v/o_proj, in_proj_qkvz, out_proj`)
`Granite-4.0-H` (`granitemoehybrid`)	Mamba-2 SSD Mixer + GQA Full-Attn + Sparse MoE & Shared MLP	Transitive $X \rightsquigarrow A \rightsquigarrow C$ Attn KV mount (98.5% agree); Mamba-2 state $\mathbf{h}_k$ skips mounts	(`d_conv` $- 1$) ancestor conv tokens; Mamba-2 SSD mapped to chunked linear attention (`fla`)	5 modules: `q_proj, k_proj, v_proj, o_proj` + Mamba `out_proj` ($H \cdot P \to D$)

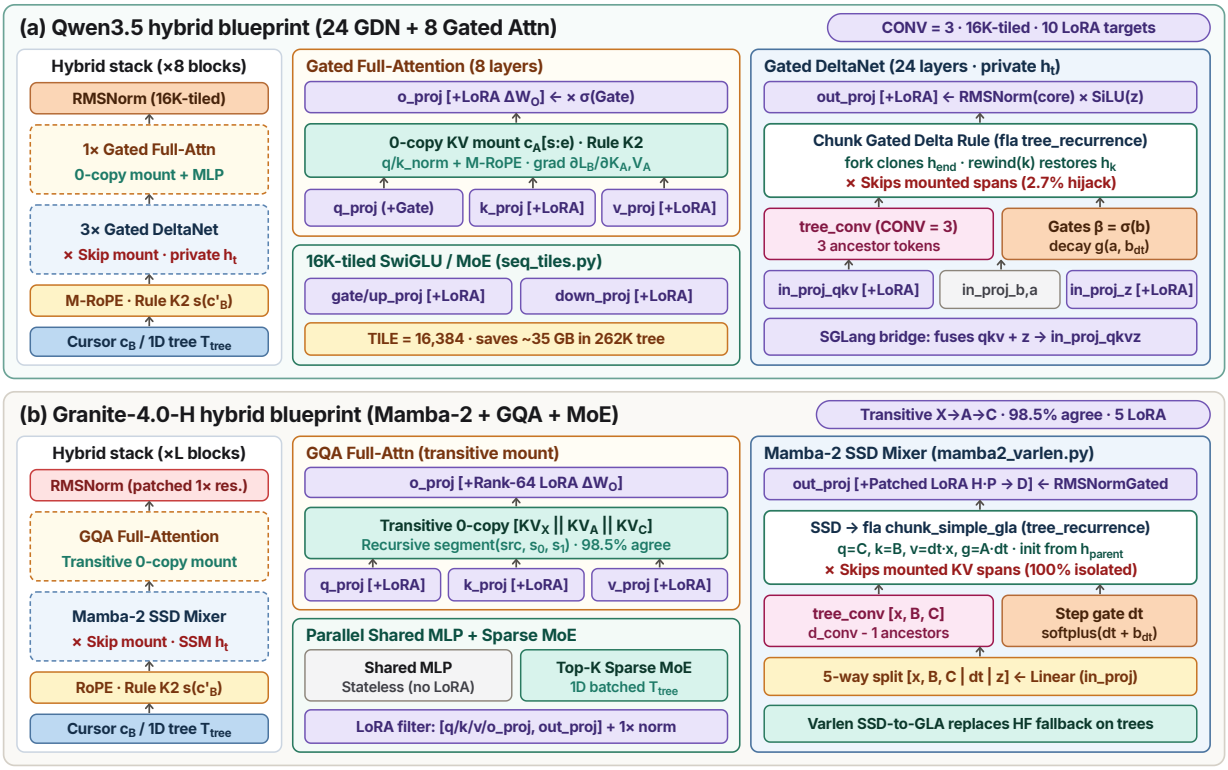


Figure 12: Octopus integration blueprints for hybrid recurrent-attention models (Table 8). **(a) Qwen3.5 (Dense & MoE Hybrid):** All 8 Gated Attention layers zero-copy mount sibling KV pages $c_A[s:e)$ under Rule K2 (Equation (2)), while all 24 GDN layers gather 3 ancestor convolution tokens and advance $\mathbf{h}_t$ along branch tokens via chunked GDN recurrence (skipping mounted spans, 2.7% hijack), paired with 16K activation tiling and fused `in_proj_qkvz` LoRA. **(b) Granite-4.0-H (Mamba-2 Hybrid):** Maps Mamba-2 SSD to chunked linear attention, supports transitive $X \rightsquigarrow A \rightsquigarrow C$ mounts (98.5% token agreement), and adapts Mamba `out_proj` and attention projections.